

Assignment to detect blackhole at the center of galaxies with help of computer vision

Step 1

Choose any one of the galaxy from galaxy centre data and write down its literature survey in 5 points.

Mention:

1. About galaxy
2. Galactic lat long
3. What is special about this galaxy
4. What has been already researched about it
5. Make your hypothesis (eg. presence of black hole, galaxy is spiral, etc.), assumption based on literature survey

Example. presence of black hole

(This is a theoritical question)

MESSIER 99 - NGC 4254

* Messier 99 or M99, also known as NGC 4254, is a grand design spiral galaxy in the northern constellation Coma Berenices approximately 15,000,000 parsecs (49,000,000 light-years) from the Milky Way. It was discovered by Pierre Méchain on 17 March 1781. The discovery was then reported to Charles Messier, who included the object in the Messier Catalogue of comet-like objects. It was one of the first galaxies in which a spiral pattern was seen. This pattern was first identified by Lord Rosse in the spring of 1846.

* This galaxy has a morphological classification of SA(s)c, indicating a pure spiral shape with loosely wound arms. It has a peculiar shape with one normal looking arm and an extended arm that is less tightly wound. The galaxy is inclined by 42° to the line-of-sight with a major axis position angle of 68°. Four supernovae have been observed in this galaxy: SN 1967H (type II), 1972Q, 1986I (type II), and 2014L (type Ic).

* A bridge of neutral hydrogen gas links NGC 4254 with VIRGOHI21, an HI region and a possible dark galaxy. The gravity from the latter may have distorted M99 and drawn out the gas bridge, as the two galaxy-sized objects may have had a close encounter before parting greatly. However, VIRGOHI21 may instead be tidal debris from an interaction with the lenticular galaxy NGC 4262 some 280 million years ago. It is expected that the drawn out arm will relax to match the normal arm once the encounter is over.

* This Hubble image, combining visible, infrared and ultraviolet observations, shows a detailed view of the spiral arms on one side of the galaxy M99. This galaxy is called a grand-design spiral, with long, large and clearly defined spiral arms — giving it a structure somewhat similar to our home galaxy, the Milky Way.

* M99 was discovered by the French astronomer Pierre Méchain in 1781. The galaxy is located 55 million light-years from Earth in the constellation Coma Berenices and has an apparent magnitude of 10.4. It can be seen using a moderately sized telescope most easily in May.

* A number of unexplained phenomena in M99 have been studied by astronomers. Among these is the nature of PTF 10fqs — one of the brighter stars in this image and visible as a yellow-orange star in the top-left corner. What is unusual about PTF 10fqs is that it has so far defied classification. When it was discovered in April 2010, it appeared brighter than a nova (caused by an eruption on a star's surface) but fainter than a supernova (the explosion that marks the end of life for a large star). Scientists have offered a number of possible explanations, including the intriguing suggestion that the star's increased brightness could have been caused by a giant planet plunging into its parent star.

* This Hubble image was made in June 2010, during the period when the outburst from PTF 10fqs was fading, so the star's location could be pinpointed with great precision. These measurements will allow other telescopes to home in on the star in future, even when the afterglow of the outburst has faded away completely.

* Constellation

Coma Berenices

* Right ascension

12h 18m 49.625s

* Declination

+14° 24' 59.36"

* Redshift

0.008029

* Helio radial velocity

2,404 km/s

* Distance

45.2 Mly (13.87 Mpc)

* Group or cluster

Virgo Cluster

* Apparent magnitude (V)

10.4

* Type

SA(s)c

* Apparent size (V)

5.4' × 4.7

Step 2

Do the exploratory data analysis on the data and write 5 points of inference from the EDA performed by you.

```
In [ ]: !pip install astropy
        !pip install specutils
```

```
Looking in indexes: https://pypi.org/simple, https://us-
python.pkg.dev/colab-wheels/public/simple/
Requirement already satisfied: astropy in
/usr/local/lib/python3.7/dist-packages (4.3.1)
Requirement already satisfied: numpy>=1.17 in
/usr/local/lib/python3.7/dist-packages (from astropy) (1.21.6)
Requirement already satisfied: importlib-metadata in
/usr/local/lib/python3.7/dist-packages (from astropy) (4.12.0)
Requirement already satisfied: pyerfa>=1.7.3 in
/usr/local/lib/python3.7/dist-packages (from astropy) (2.0.0.1)
Requirement already satisfied: typing-extensions>=3.6.4 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata-
>astropy) (4.1.1)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata-
>astropy) (3.8.1)
Looking in indexes: https://pypi.org/simple, https://us-
python.pkg.dev/colab-wheels/public/simple/
Requirement already satisfied: specutils in
/usr/local/lib/python3.7/dist-packages (1.8.0)
Requirement already satisfied: gwcs>=0.17.0 in
/usr/local/lib/python3.7/dist-packages (from specutils) (0.18.1)
Requirement already satisfied: astropy>=4.1 in
/usr/local/lib/python3.7/dist-packages (from specutils) (4.3.1)
Requirement already satisfied: scipy in
/usr/local/lib/python3.7/dist-packages (from specutils) (1.7.3)
Requirement already satisfied: ndcube>=2.0 in
/usr/local/lib/python3.7/dist-packages (from specutils) (2.0.2)
Requirement already satisfied: asdf>=2.5 in
/usr/local/lib/python3.7/dist-packages (from specutils) (2.12.0)
Requirement already satisfied: jsonschema>=4.0.1 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(4.3.3)
Requirement already satisfied: packaging>=16.0 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(21.3)
Requirement already satisfied: six in /usr/local/lib/python3.7/dist-
packages (from asdf>=2.5->specutils) (1.15.0)
Requirement already satisfied: pyyaml>=3.10 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(6.0)
Requirement already satisfied: asdf-transform-schemas>=0.2.2 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(0.2.2)
Requirement already satisfied: jmespath>=0.6.2 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(1.0.1)
Requirement already satisfied: asdf-standard>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(1.0.1)
Requirement already satisfied: numpy>=1.10 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(1.21.6)
Requirement already satisfied: semantic-version>=2.8 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(2.10.0)
Requirement already satisfied: importlib-resources>=3 in
/usr/local/lib/python3.7/dist-packages (from asdf>=2.5->specutils)
(5.9.0)
Requirement already satisfied: importlib-metadata in
/usr/local/lib/python3.7/dist-packages (from astropy>=4.1->specutils)
(4.12.0)
Requirement already satisfied: pyerfa>=1.7.3 in
/usr/local/lib/python3.7/dist-packages (from astropy>=4.1->specutils)
(2.0.0.1)
Requirement already satisfied: asdf-astropy>=0.2.0 in
```

```
/usr/local/lib/python3.7/dist-packages (from gwcs>=0.17.0->specutils)
(0.2.0)
Requirement already satisfied: asdf-wcs-schemas in
/usr/local/lib/python3.7/dist-packages (from gwcs>=0.17.0->specutils)
(0.1.1)
Requirement already satisfied: asdf-coordinates-schemas in
/usr/local/lib/python3.7/dist-packages (from asdf-astropy>=0.2.0-
>gwcs>=0.17.0->specutils) (0.1.0)
Requirement already satisfied: zipp>=3.1.0 in
/usr/local/lib/python3.7/dist-packages (from importlib-resources>=3-
>asdf>=2.5->specutils) (3.8.1)
Requirement already satisfied:
pysistent!=0.17.0,!0.17.1,!0.17.2,>=0.14.0 in
/usr/local/lib/python3.7/dist-packages (from jsonschema>=4.0.1-
>asdf>=2.5->specutils) (0.18.1)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from jsonschema>=4.0.1-
>asdf>=2.5->specutils) (4.1.1)
Requirement already satisfied: attrs>=17.4.0 in
/usr/local/lib/python3.7/dist-packages (from jsonschema>=4.0.1-
>asdf>=2.5->specutils) (22.1.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in
/usr/local/lib/python3.7/dist-packages (from packaging>=16.0-
>asdf>=2.5->specutils) (3.0.9)
```

```
In [ ]: # importing necessary libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib as mpl
import matplotlib.cm as mpltcm
from mpl_toolkits.mplot3d import Axes3D
import random
import time
```

```
In [ ]: from astropy.visualization import astropy_mpl_style
from astropy.utils.data import get_pkg_data_filename
from astropy.io import fits
import astropy
from astropy import units as u
import astropy.wcs as fitswcs
from astropy.cosmology import WMAP9 as Cosmology
```

```
In [ ]: # Ignoring Warnings

from warnings import filterwarnings

filterwarnings("ignore",category=DeprecationWarning)
filterwarnings("ignore", category=FutureWarning)
filterwarnings("ignore", category=UserWarning)
filterwarnings("ignore", category=RuntimeWarning)
```

```
In [ ]: # Getting data from drive
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Drive already mounted at /content/gdrive; to attempt to forcibly
remount, call drive.mount("/content/gdrive", force_remount=True).
```

```
In [ ]: ! unzip /content/gdrive/MyDrive/poojan/GALAXIES_CENTER.zip
```

```
In [ ]: # Reading fits data
NEAR_INFRARED_PATH = "/content/M99/NEAR_INFRARED/n3zb2tboq_cal.fits"
HST_OPTICAL_PATH =
"/content/M99/OPTICAL/hst_08597_40_wfpc2_f606w_pc_drz.fits"
XMM_NEWTON_PATH =
"/content/M99/XMM_Newton_Soft_Xray/P0147610101EPX0003COLIM8000.FTZ"
```

```
In [ ]: # opening fits data
NI_OPEN = fits.open(NEAR_INFRARED_PATH)
HST_OPEN = fits.open(HST_OPTICAL_PATH)
XMM_NEWTON_OPEN = fits.open(XMM_NEWTON_PATH)
```

```
In [ ]: OPEN_LIST = [NI_OPEN,HST_OPEN,XMM_NEWTON_OPEN]
OPEN_NAME_LIST = ["NEAR_INFRARED","HST","NEWTON"]
```

```
In [ ]: for x_open,x_name in zip(OPEN_LIST,OPEN_NAME_LIST):
        print(x_name)
        print("---"*10)
        print(x_open.info())
        print("\n")
```

```
NEAR_INFRARED
-----
Filename: /content/M99/NEAR_INFRARED/n3zb2tboq_cal.fits
No.    Name      Ver    Type      Cards  Dimensions  Format
  0  PRIMARY      1 PrimaryHDU    309      ()
  1  SCI          1 ImageHDU     143  (256, 256)  float32
  2  ERR          1 ImageHDU      71  (256, 256)  float32
  3  DQ           1 ImageHDU      71  (256, 256)  int16
  4  SAMP         1 ImageHDU      71  (256, 256)  int16
  5  TIME         1 ImageHDU      71  (256, 256)  float32
None
```

```
HST
-----
Filename: /content/M99/OPTICAL/hst_08597_40_wfpc2_f606w_pc_drz.fits
No.    Name      Ver    Type      Cards  Dimensions  Format
  0  PRIMARY      1 PrimaryHDU    334      ()
  1  SCI          1 ImageHDU     110  (950, 950)  float32
  2  WHT          1 ImageHDU     130  (950, 950)  float32
  3  CTX          1 ImageHDU     129  (950, 950)  int32
None
```

```
NEWTON
-----
Filename:
/content/M99/XMM_Newton_Soft_Xray/P0147610101EPX0003COLIM8000.FTZ
No.    Name      Ver    Type      Cards  Dimensions  Format
  0  PRIMARY      1 PrimaryHDU     56  (648, 648, 3)  float32
  1  RAWIMAGE     1 ImageHDU      34  (648, 648, 3)  float32
  2  FWCIMAGE     1 ImageHDU      34  (648, 648, 3)  float32
  3  OOTIMAGE     1 ImageHDU      34  (648, 648, 3)  float32
None
```

```
In [ ]: NI_PRIMARY_OPERATION = NI_OPEN[0].header
```

In []:

```
# Finding some parameters about the galaxy from the provided data
print(NI_PRIMARY_OPERATION.comments["TARGNAME"] + "
",NI_PRIMARY_OPERATION["TARGNAME"])
print(NI_PRIMARY_OPERATION.comments["RA_TARG"] + "
",NI_PRIMARY_OPERATION["RA_TARG"])
print(NI_PRIMARY_OPERATION.comments["DEC_TARG"] + "
",NI_PRIMARY_OPERATION["DEC_TARG"])
print(NI_PRIMARY_OPERATION.comments["GAL_LAT"] + "
",NI_PRIMARY_OPERATION["GAL_LAT"])
print(NI_PRIMARY_OPERATION.comments["GAL_LONG"] + "
",NI_PRIMARY_OPERATION["GAL_LONG"])
```

```
proposer's target name   NGC4254
right ascension of the target (deg) (J2000)  184.7059583333
declination of the target (deg) (J2000)  14.41652777778
galactic latitude of the target (deg) (J2000)  75.189472
galactic longitude of the target (deg) (J2000)  270.428235
```

```
In [ ]: print(NI_PRIMARY_OPERATION.comments["DEC_TARG"] + "
",NI_PRIMARY_OPERATION["DEC_TARG"])
print(NI_PRIMARY_OPERATION.comments["ECL_LAT"] + "
",NI_PRIMARY_OPERATION["ECL_LAT"])
print(NI_PRIMARY_OPERATION.comments["ECL_LONG"] + "
",NI_PRIMARY_OPERATION["ECL_LONG"])
print(NI_PRIMARY_OPERATION.comments["SUNANGLE"] + "
",NI_PRIMARY_OPERATION["SUNANGLE"])
print(NI_PRIMARY_OPERATION.comments["MOONANGL"] + "
",NI_PRIMARY_OPERATION["MOONANGL"])
print(NI_PRIMARY_OPERATION.comments["DATE-OBS"] + "
",NI_PRIMARY_OPERATION["DATE-OBS"])
print(NI_PRIMARY_OPERATION.comments["TIME-OBS"] + "
",NI_PRIMARY_OPERATION["TIME-OBS"])
print(NI_PRIMARY_OPERATION.comments["EXPSTART"] + "
",NI_PRIMARY_OPERATION["EXPSTART"])
print(NI_PRIMARY_OPERATION.comments["EXPEND"] + "
",NI_PRIMARY_OPERATION["EXPEND"])
print(NI_PRIMARY_OPERATION.comments["EXPTIME"] + "
",NI_PRIMARY_OPERATION["EXPTIME"])
print(NI_PRIMARY_OPERATION.comments["PATTERN1"] + "
",NI_PRIMARY_OPERATION["PATTERN1"])
print(NI_PRIMARY_OPERATION.comments["P1_SHAPE"] + "
",NI_PRIMARY_OPERATION["P1_SHAPE"])
print(NI_PRIMARY_OPERATION.comments["P1_PURPS"] + "
",NI_PRIMARY_OPERATION["P1_PURPS"])
print(NI_PRIMARY_OPERATION.comments["P1_NPTS"] + " ", "%.30f" %
NI_PRIMARY_OPERATION["P1_NPTS"])
print(NI_PRIMARY_OPERATION.comments["P1_PSPAC"] + " ", "%.30f" %
NI_PRIMARY_OPERATION["P1_PSPAC"])
print(NI_PRIMARY_OPERATION.comments["P1_LSPAC"] + " ", "%.30f" %
NI_PRIMARY_OPERATION["P1_LSPAC"])
print(NI_PRIMARY_OPERATION.comments["PHOTFLAM"] + " ", "%.30f" %
NI_PRIMARY_OPERATION["PHOTFLAM"])
print(NI_PRIMARY_OPERATION.comments["PHOTMODE"] + "
",NI_PRIMARY_OPERATION["PHOTMODE"])
print(NI_PRIMARY_OPERATION.comments["PHOTFNU"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["PHOTFNU"])
print(NI_PRIMARY_OPERATION.comments["PHOTZPT"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["PHOTZPT"])
print(NI_PRIMARY_OPERATION.comments["PHOTPLAM"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["PHOTPLAM"])
print(NI_PRIMARY_OPERATION.comments["PHOTBW"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["PHOTBW"])
print(NI_PRIMARY_OPERATION.comments["ZPSCALE"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["ZPSCALE"])
print(NI_PRIMARY_OPERATION.comments["PHOTFERR"] + " ", "%.10f" %
NI_PRIMARY_OPERATION["PHOTFERR"])
```

```

declination of the target (deg) (J2000) 14.41652777778
ecliptic latitude of the target (deg) (J2000) 15.071945
ecliptic longitude of the target (deg) (J2000) 178.449199
angle between sun and V1 axis 73.460861
angle between moon and V1 axis 42.548717
UT date of start of observation (yyyy-mm-dd) 1997-07-07
UT time of start of observation (hh:mm:ss) 18:13:26
exposure start time (Modified Julian Date) 50636.75933279
exposure end time (Modified Julian Date) 50636.76748093
exposure duration (seconds)--calculated 703.9361
primary pattern type NONE
primary pattern shape LINE
primary pattern purpose MOSAIC
number of points in primary pattern 0.000000000000000000000000000000
point spacing for primary pattern (arc-sec)
0.000000000000000000000000000000
line spacing for primary pattern (arc-sec)
0.000000000000000000000000000000
inverse sensitivity (ergs/cm**2/Angstrom/DN)
0.000000000000000000000000000000

```

```
NICMOS,2,F160W,DN
inverse sensitivity (JY*sec/DN)  0.0000020304
ST magnitude system zero point (mag)  -21.1000000000
pivot wavelength of the photmode (Angstroms)  16059.9000000000
RMS bandwidth of the photmode (Angstroms)  1177.3000000000
temp dependent photometric 0-pt scale factor  1.0183918000
relat err: temp dependent photometric zero-pt  0.0150000000
```

INFERENCE

- 1. Data of Galaxy M99 is available in 3 sensors; Near Infrared(NI), Optical(HST) & Newton.
- 2. Image distribution is as follows:-
NI -> 5 images
HST -> 3 images
Newton -> 4 images
- 3. Galaxy M99 is also noted as NGC-4254 & similar other data can be found in the FITS data provided which has been extracted as shown above.
- 4. By extracting useful data such as declination angle, ecliptical latitude & longitude etc. we can get an idea about the galaxy's precise location by summarizing the available FITS data.
- 5. By reading the FITS data we can also know about the dimensions of the images as from the data shown above we can say that the images from different sensors are of different dimension so this can be very helpful while performing data analysis and visualizing the data.

Step 3

To prove your hypothesis and have better insights on data, use atleast 2 filters and view the filtered data in atleast 5 colormaps.

Write 3 points of inference, one on filter and its impact on image and other on colormap. Last point as a conclusion telling, which filter and colormap gave better insights

```
In [ ]: # Reading Images in FITS data for DATA VIZUALIZATION
NI_SCI = NI_OPEN[1].data
NI_ERR = NI_OPEN[2].data
NI_DQ = NI_OPEN[3].data
NI_SAMP = NI_OPEN[4].data
NI_TIME = NI_OPEN[5].data
HST_SCI = HST_OPEN[1].data
HST_WHT = HST_OPEN[2].data
HST_CTX = HST_OPEN[3].data
NEWTON_PRIMARY = XMM_NEWTON_OPEN[0].data
NEWTON_RAW = XMM_NEWTON_OPEN[1].data
NEWTON_FWC = XMM_NEWTON_OPEN[2].data
NEWTON_OOT = XMM_NEWTON_OPEN[3].data
```

```
In [ ]: # Loading Image Processing Libraries
import cv2
import skimage
from skimage import io, filters
from scipy import signal
```

```
In [ ]: # Viewing different color maps
        SPECTRAL_LIST = ["gray", "jet", "hot", "prism", "nipy_spectral",
                          "gist_ncar", "gist_earth", "gist_stern", "flag", "gnuplot2", 'terrain']
```

```
In [ ]: # NEAR-INFRARED IMAGES
for x_spec in SPECTRAL_LIST:

    figure,axis = plt.subplots(1,5,figsize=(25,25))

    axis[0].imshow(NI_SCI / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("NEAR INFRARED SCI" + " / "+ x_spec)
    axis[0].axis("off")

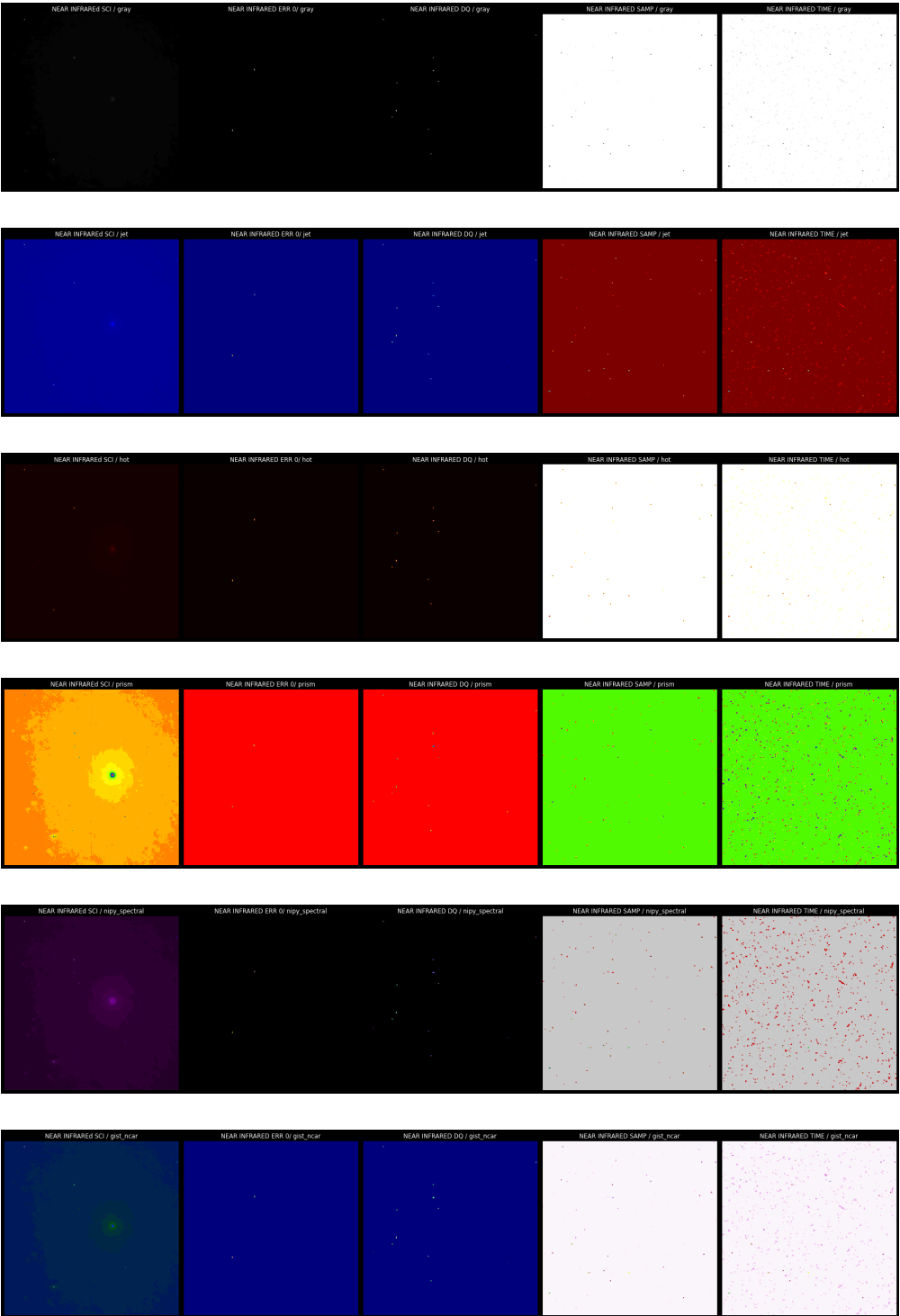
    axis[1].imshow(NI_ERR / 255.,cmap=x_spec)
    axis[1].set_title("NEAR INFRARED ERR" + " 0/ " + x_spec)
    axis[1].axis("off")

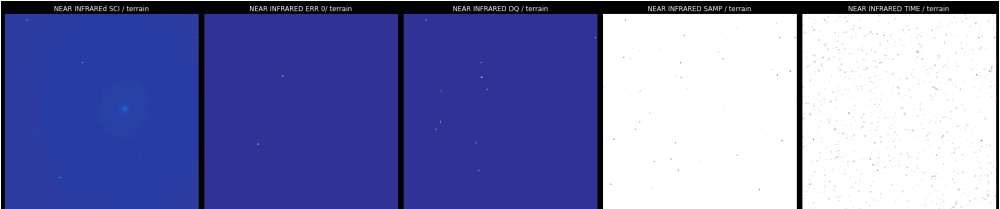
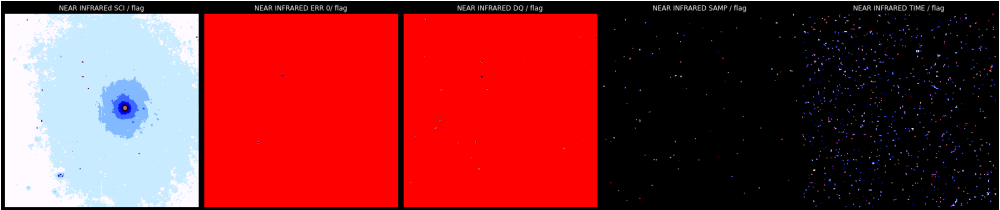
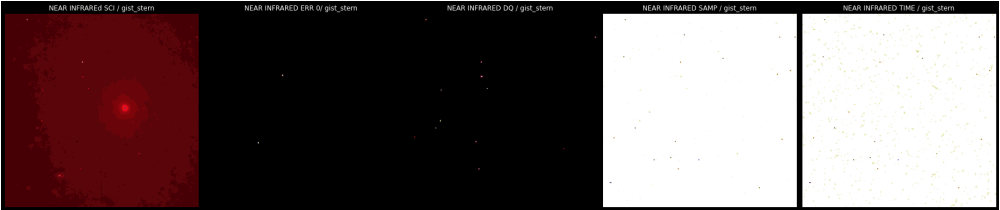
    axis[2].imshow(NI_DQ / 255.,cmap=x_spec)
    axis[2].set_title("NEAR INFRARED DQ" + " / " + x_spec)
    axis[2].axis("off")

    axis[3].imshow(NI_SAMP / 255.,cmap=x_spec)
    axis[3].set_title("NEAR INFRARED SAMP" + " / " + x_spec)
    axis[3].axis("off")

    axis[4].imshow(NI_TIME / 255.,cmap=x_spec)
    axis[4].set_title("NEAR INFRARED TIME" + " / " + x_spec)
    axis[4].axis("off")

    plt.tight_layout()
    plt.show()
```





```
In [ ]: # HST OPTICAL IMAGES
for x_spec in SPECTRAL_LIST:

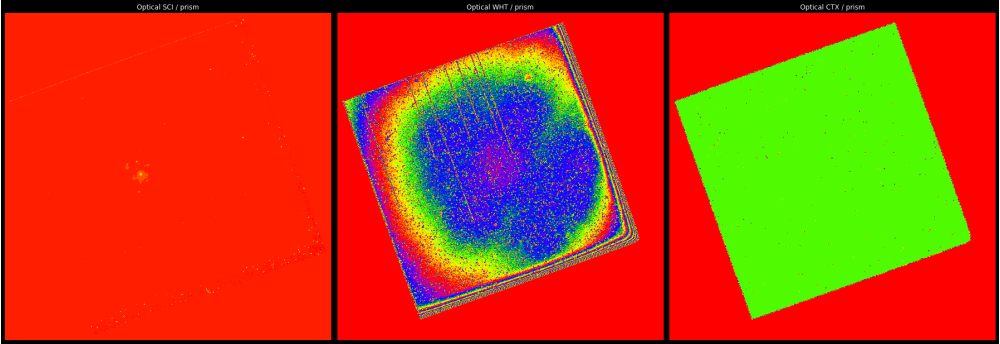
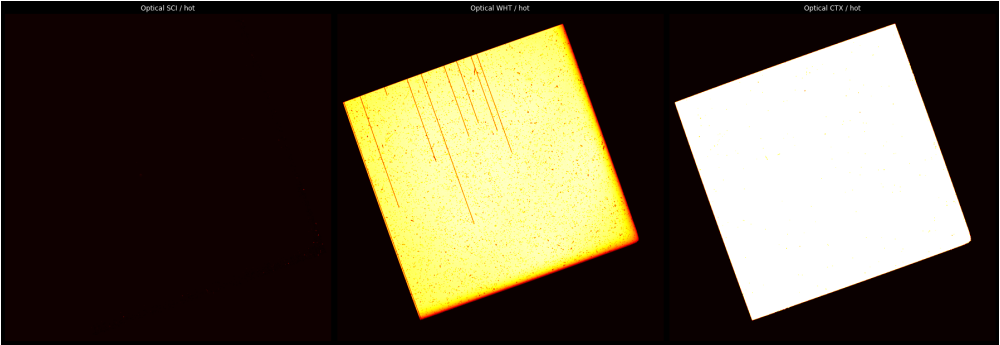
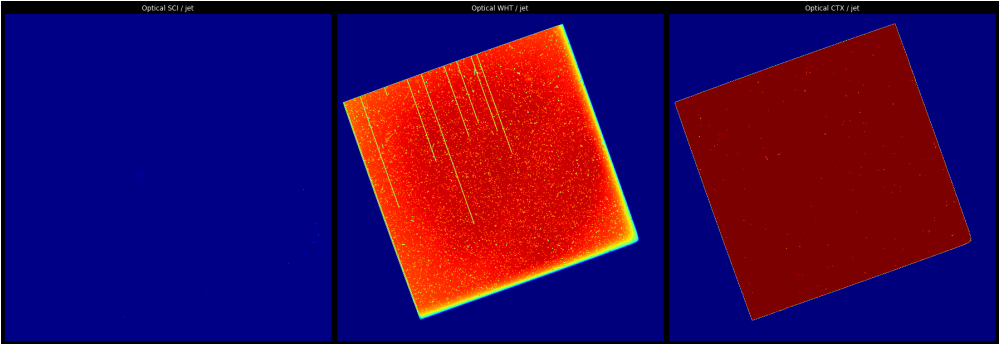
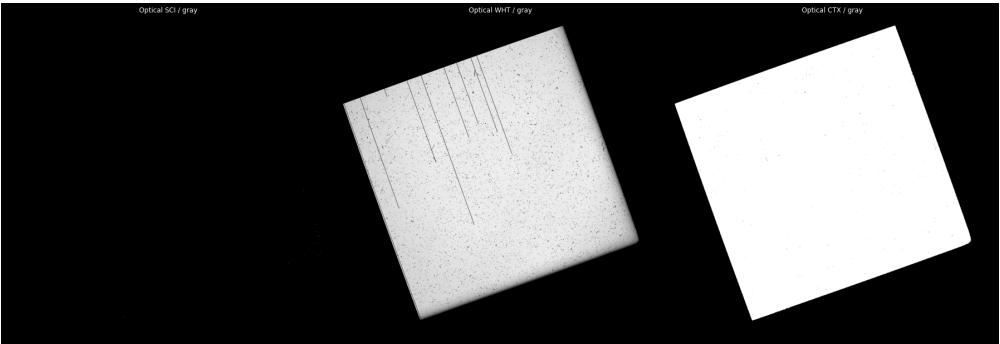
    figure,axis = plt.subplots(1,3,figsize=(25,25))

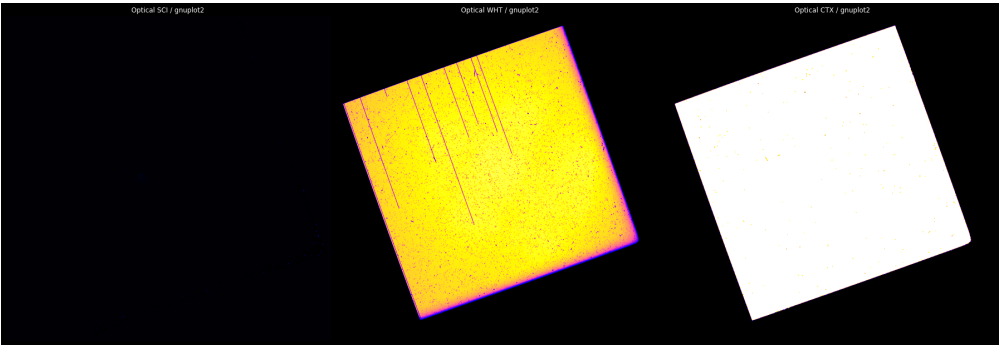
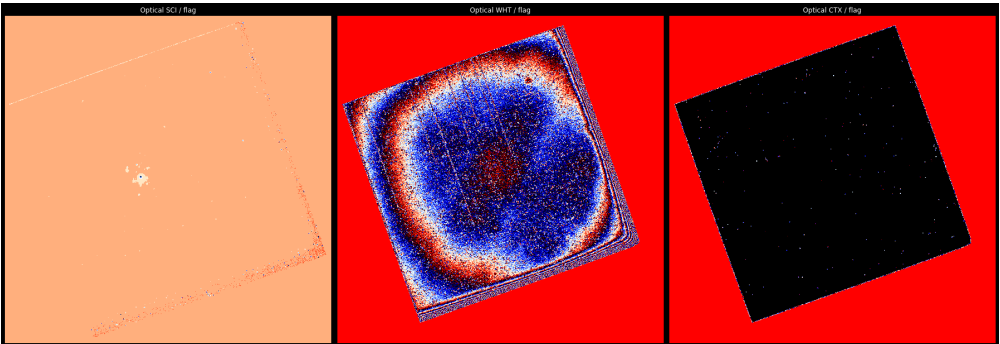
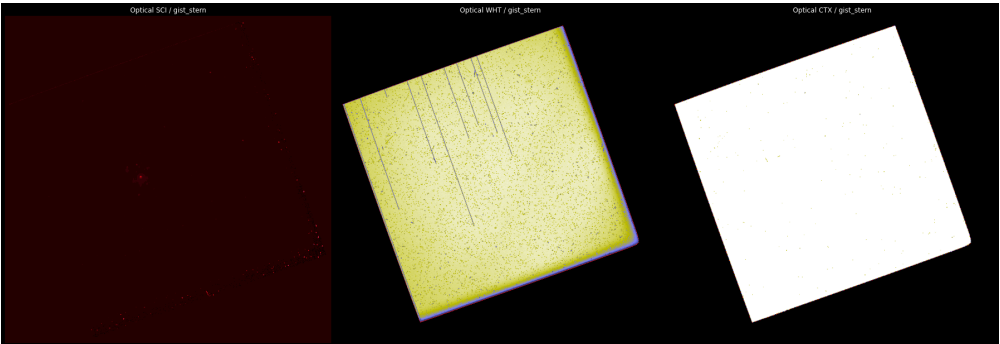
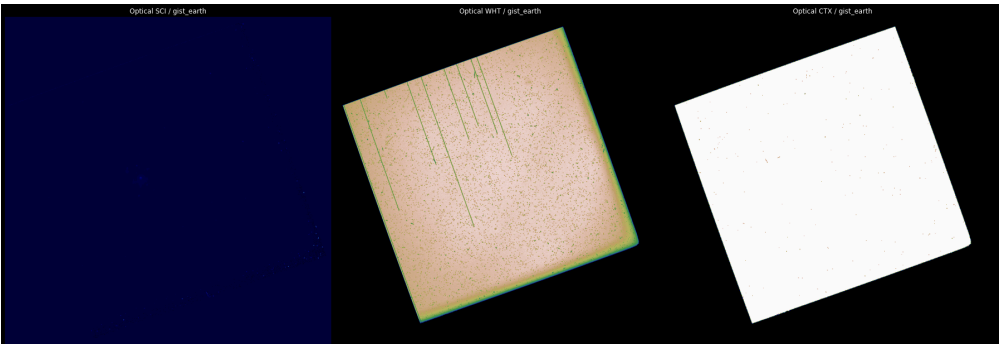
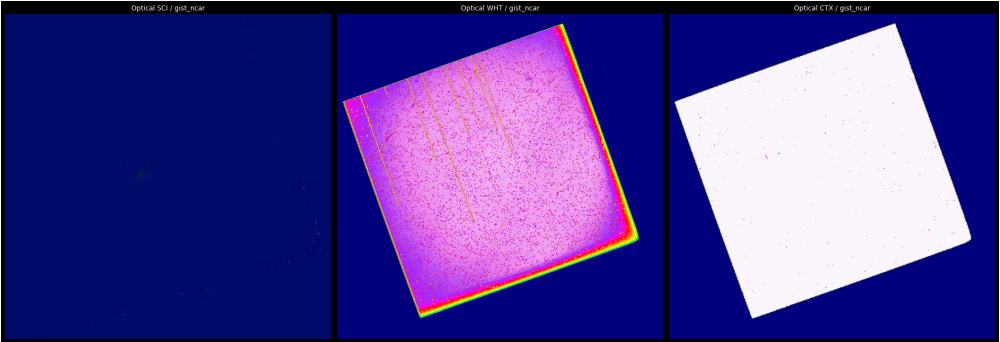
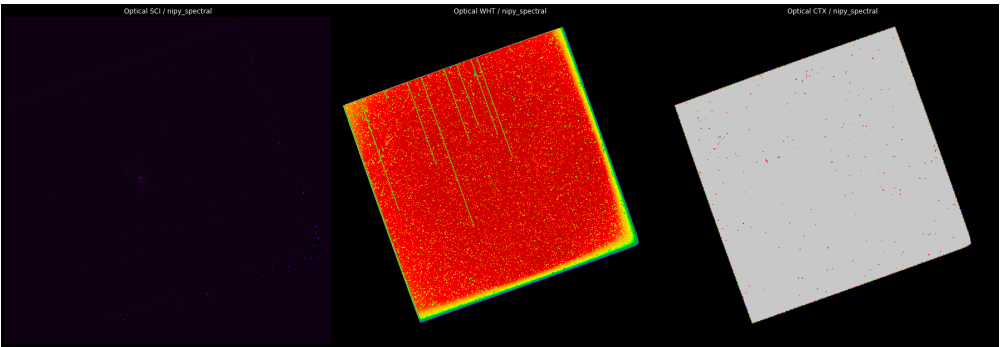
    axis[0].imshow(HST_SCI / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Optical SCI" + " / " + x_spec)
    axis[0].axis("off")

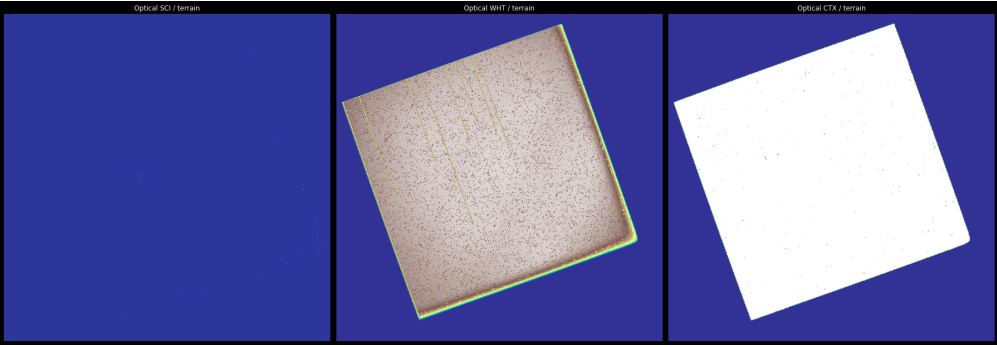
    axis[1].imshow(HST_WHT / 255.,cmap=x_spec)
    axis[1].set_title("Optical WHT" + " / " + x_spec)
    axis[1].axis("off")

    axis[2].imshow(HST_CTX / 255.,cmap=x_spec)
    axis[2].set_title("Optical CTX" + " / " + x_spec)
    axis[2].axis("off")

    plt.tight_layout()
    plt.show()
```







```
In [ ]: # XMM NEWTON PRIMARY IMAGES
for x_spec in SPECTRAL_LIST:

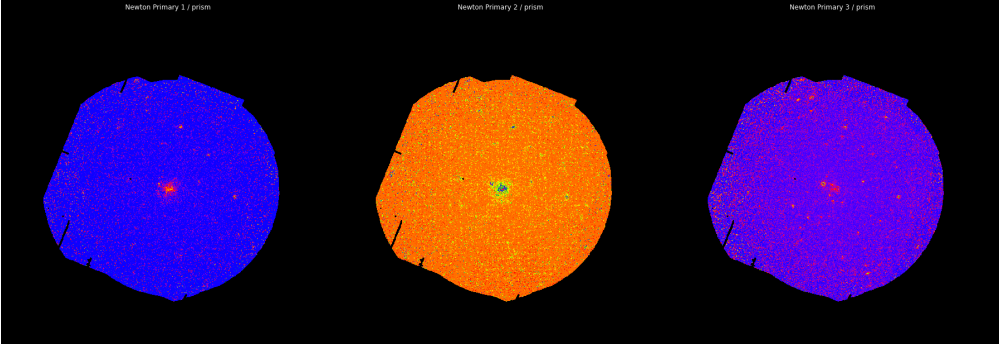
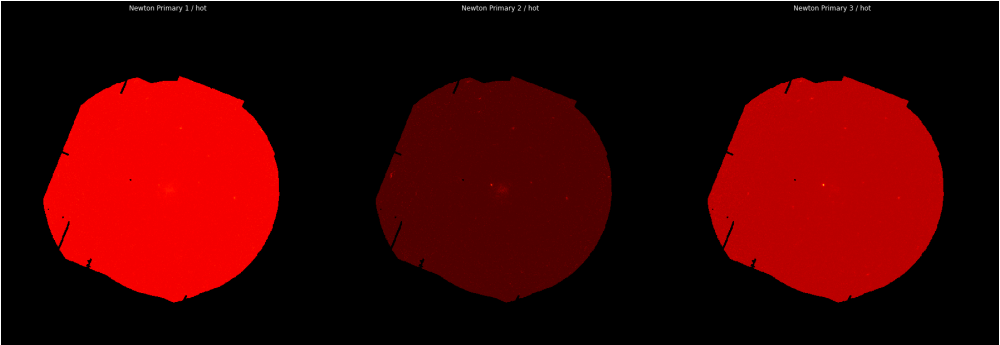
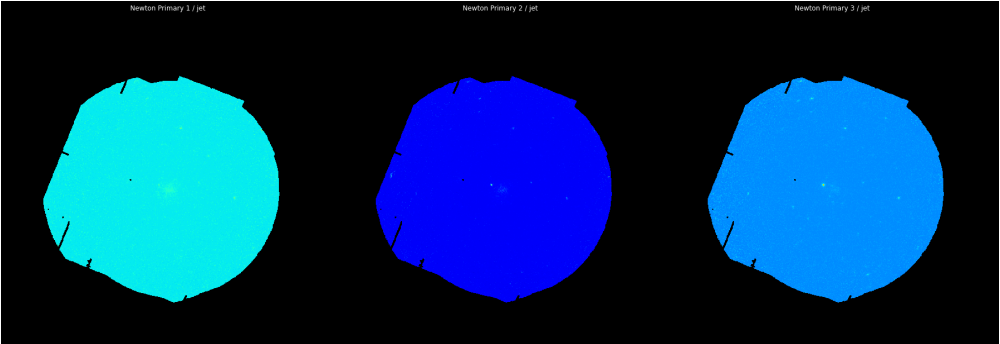
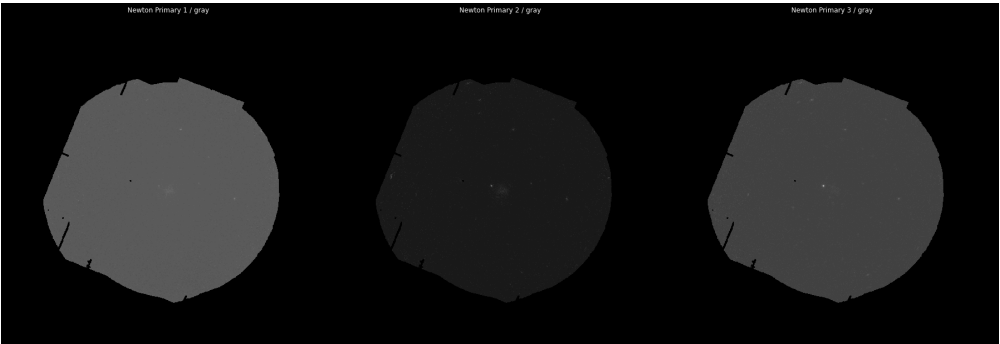
    figure,axis = plt.subplots(1,3,figsize=(25,25))

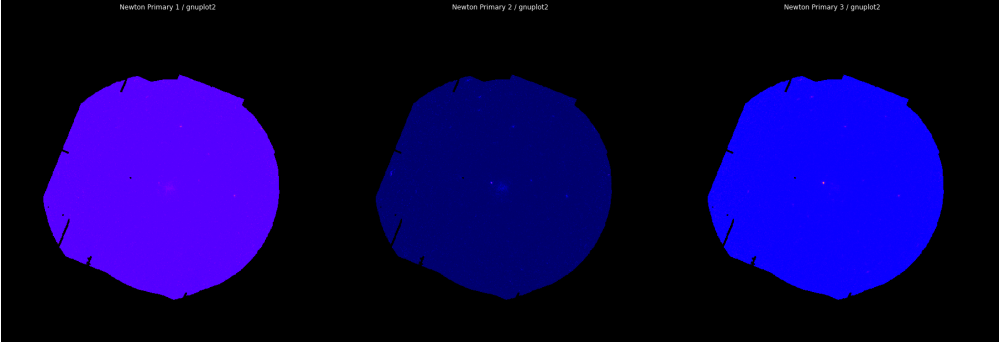
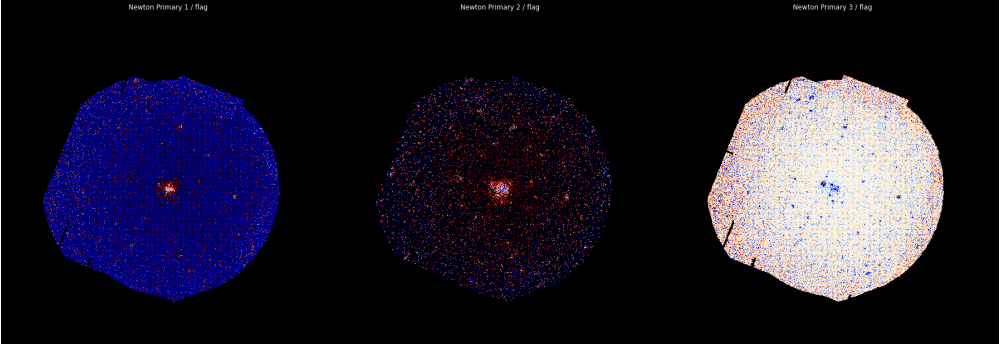
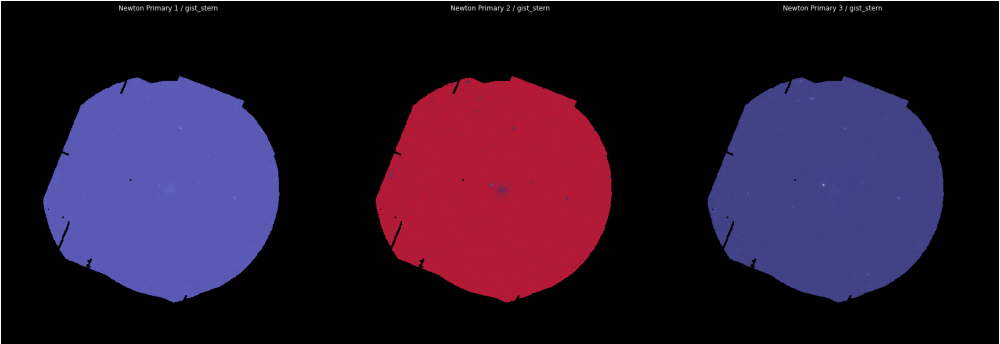
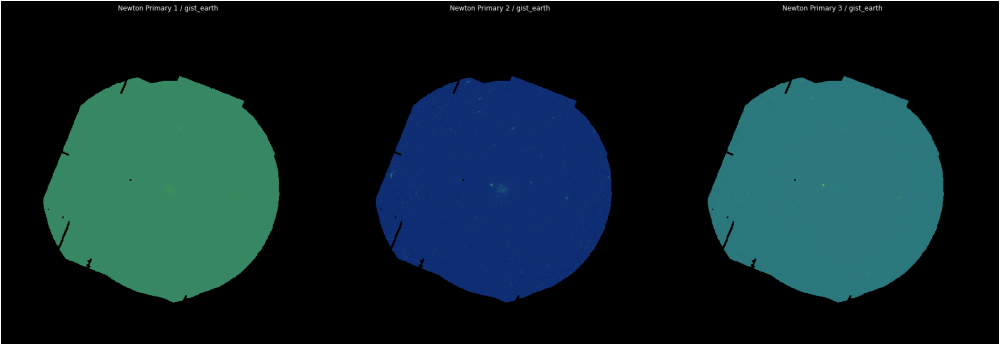
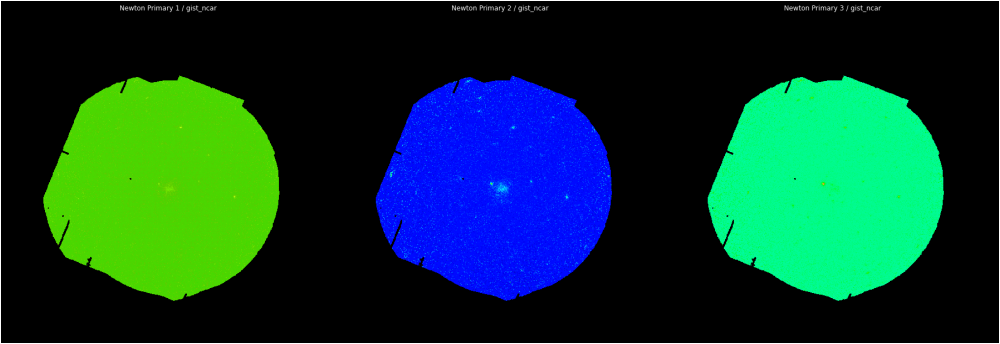
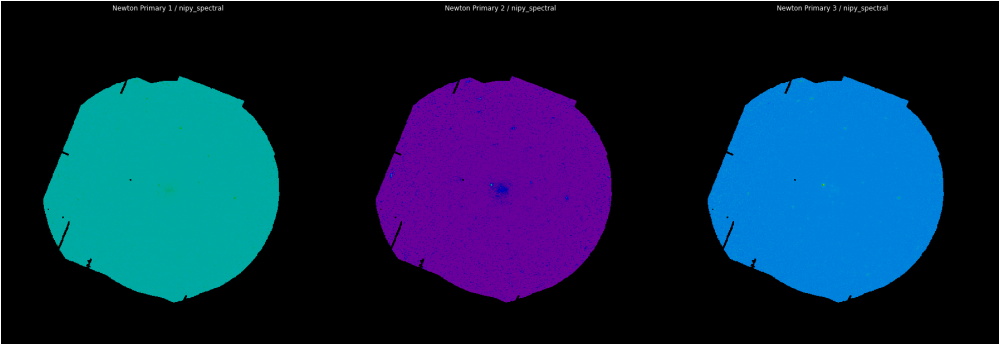
    axis[0].imshow(NEWTON_PRIMARY[0] / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Newton Primary 1" + " / " + x_spec)
    axis[0].axis("off")

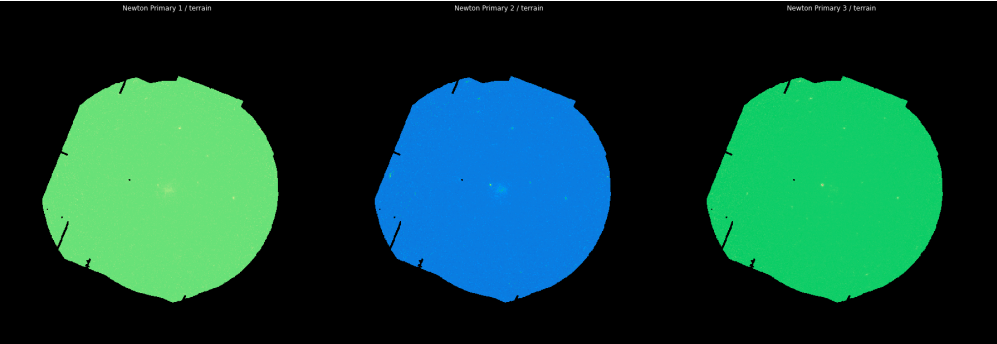
    axis[1].imshow(NEWTON_PRIMARY[1] / 255.,cmap=x_spec)
    axis[1].set_title("Newton Primary 2" + " / " + x_spec)
    axis[1].axis("off")

    axis[2].imshow(NEWTON_PRIMARY[2] / 255.,cmap=x_spec)
    axis[2].set_title("Newton Primary 3" + " / " + x_spec)
    axis[2].axis("off")

    plt.tight_layout()
    plt.show()
```







```
In [ ]: # XMM NEWTON RAW IMAGES
for x_spec in SPECTRAL_LIST:

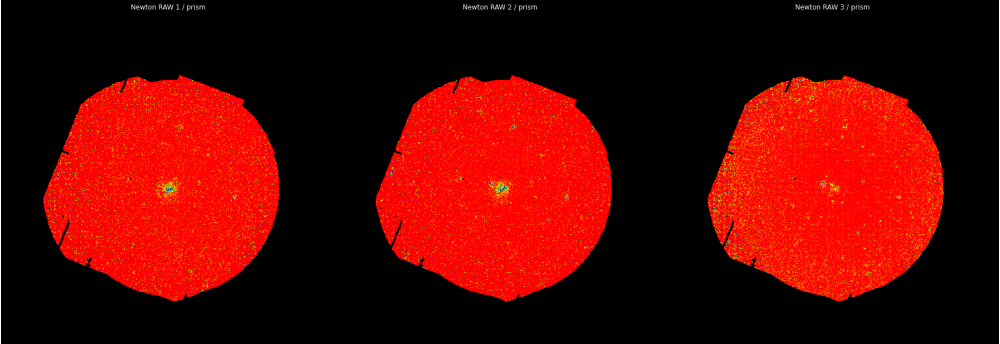
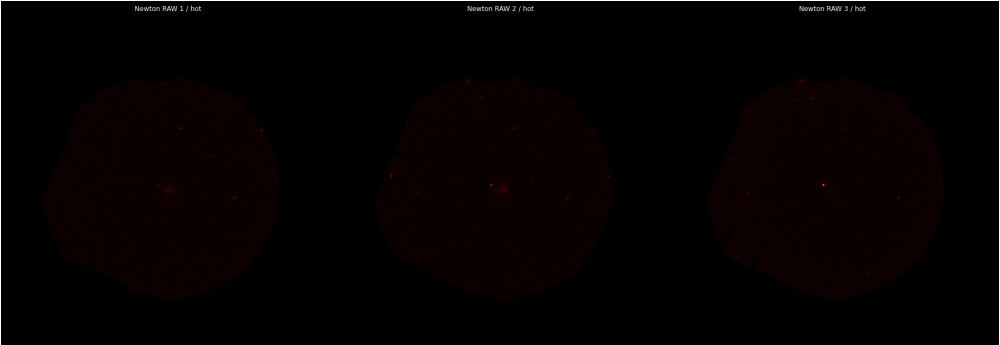
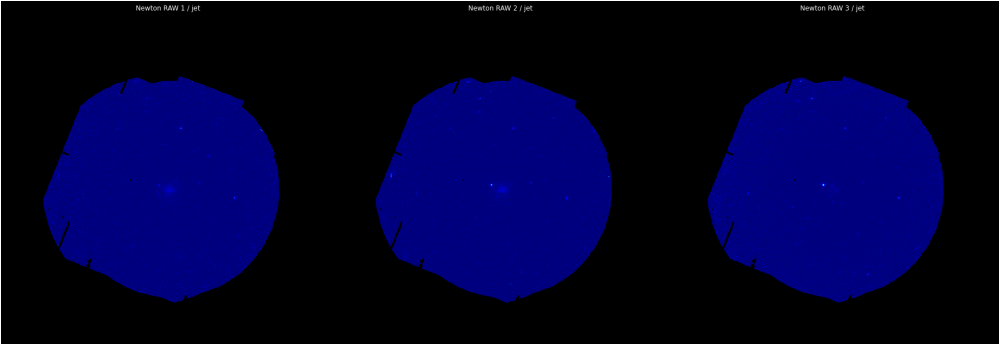
    figure,axis = plt.subplots(1,3,figsize=(25,25))

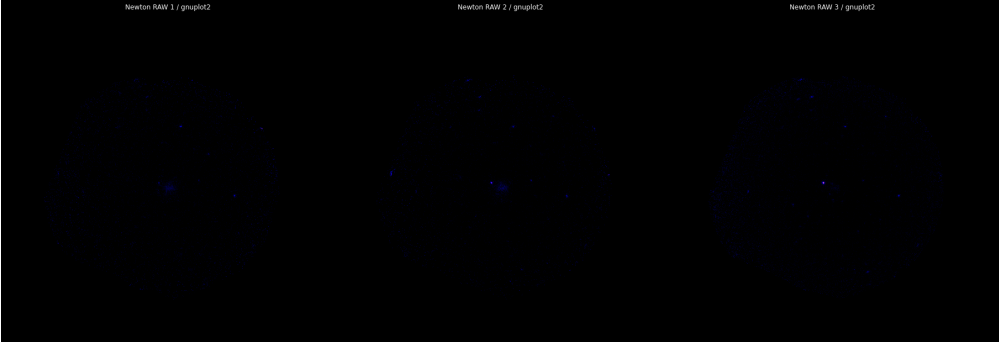
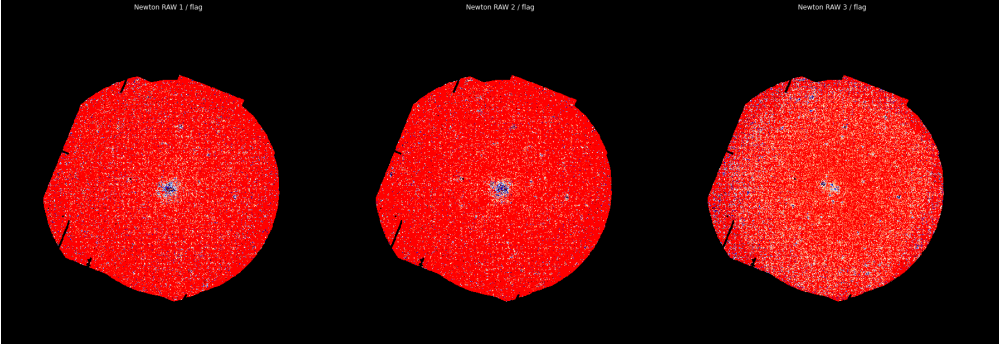
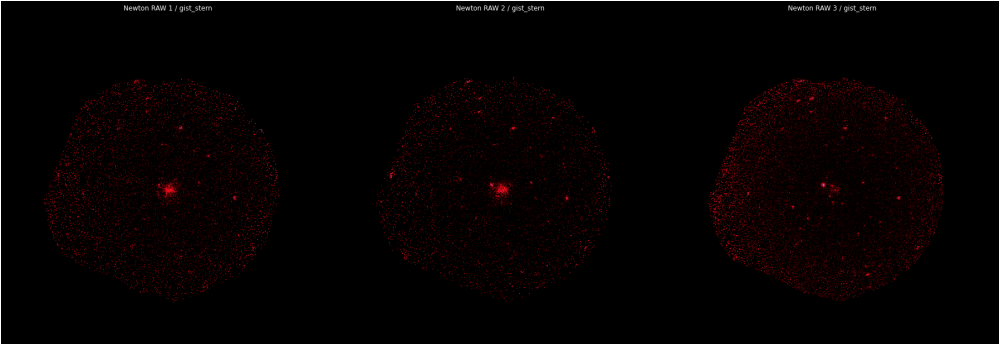
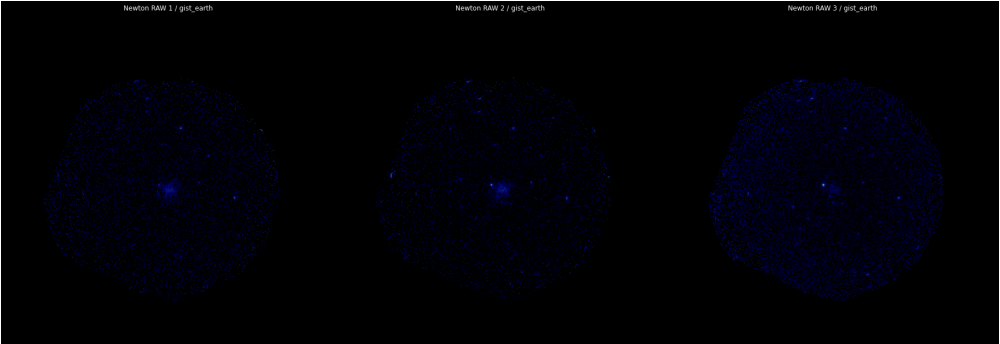
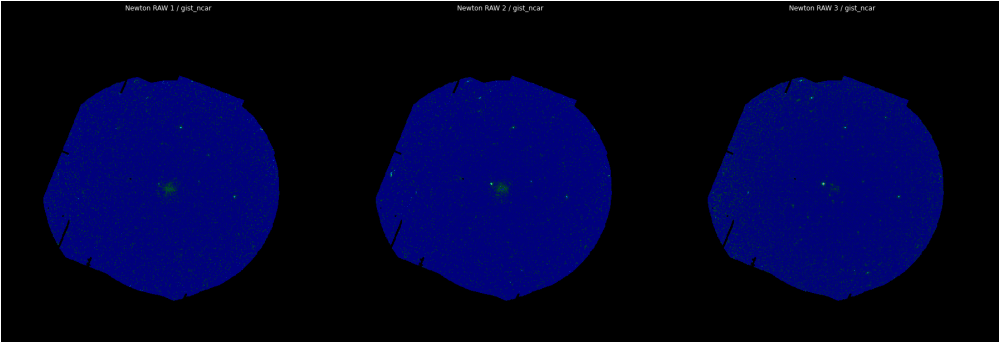
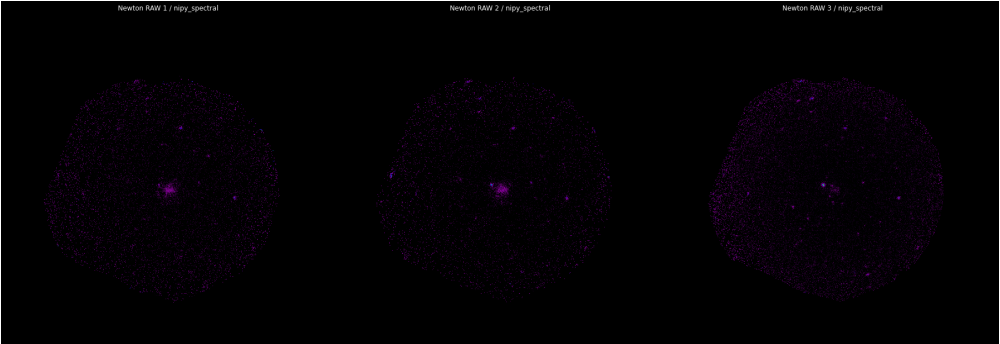
    axis[0].imshow(NEWTON_RAW[0] / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Newton RAW 1" + " / " + x_spec)
    axis[0].axis("off")

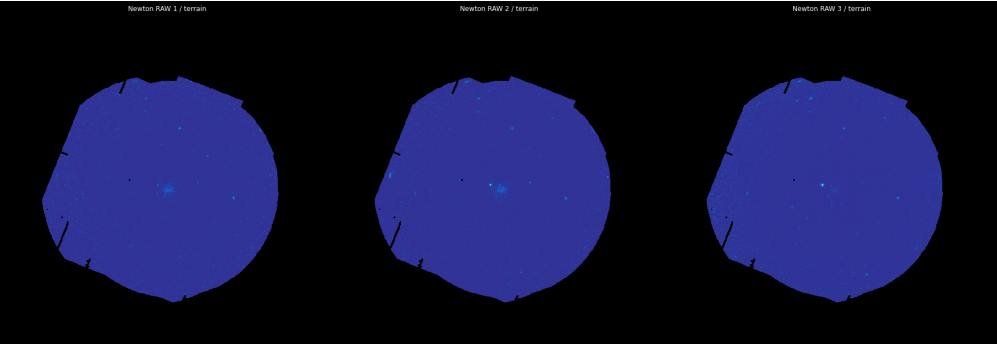
    axis[1].imshow(NEWTON_RAW[1] / 255.,cmap=x_spec)
    axis[1].set_title("Newton RAW 2" + " / " + x_spec)
    axis[1].axis("off")

    axis[2].imshow(NEWTON_RAW[2] / 255.,cmap=x_spec)
    axis[2].set_title("Newton RAW 3" + " / " + x_spec)
    axis[2].axis("off")

    plt.tight_layout()
    plt.show()
```







```
In [ ]: # XMM NEWTON FWC IMAGES
for x_spec in SPECTRAL_LIST:

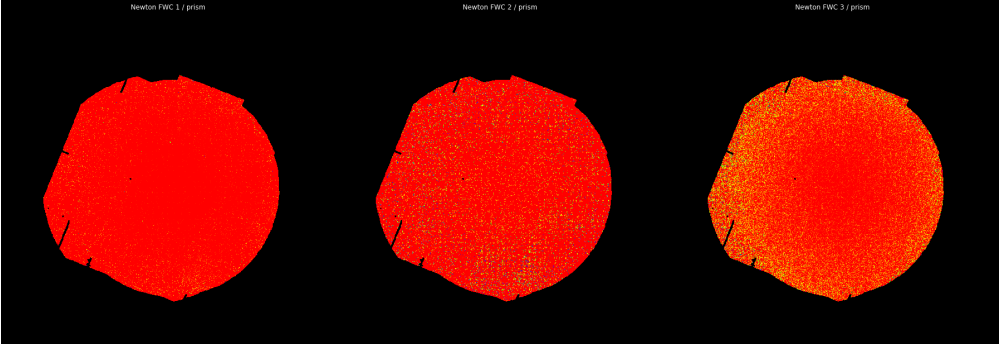
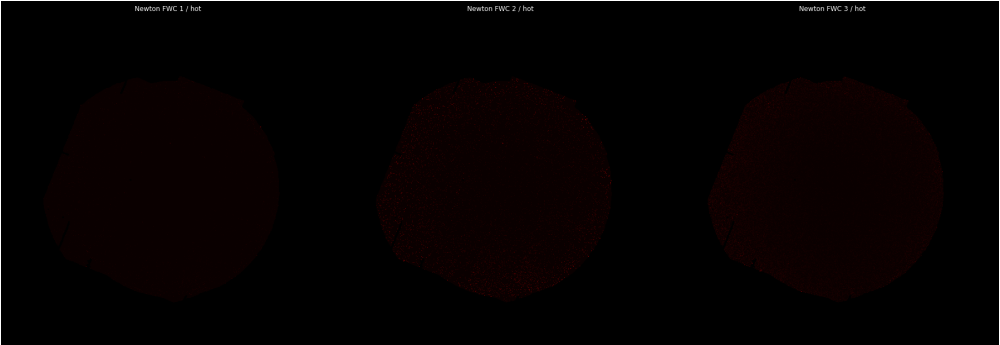
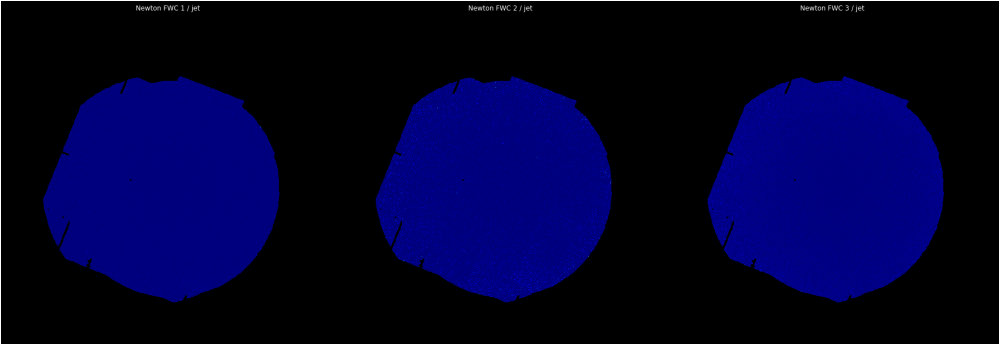
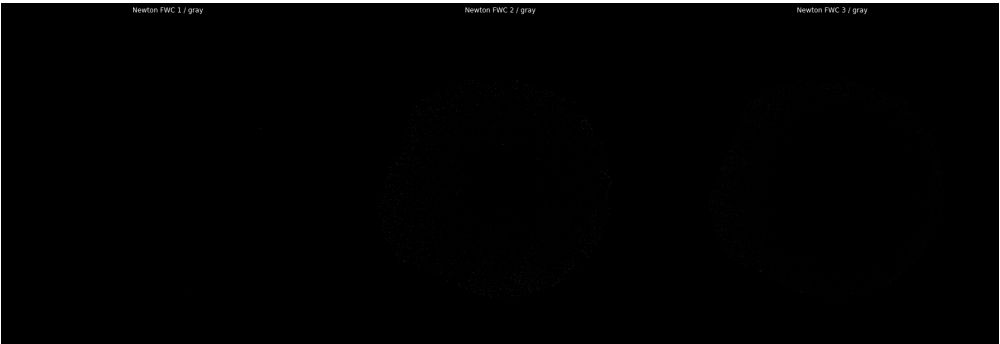
    figure,axis = plt.subplots(1,3,figsize=(25,25))

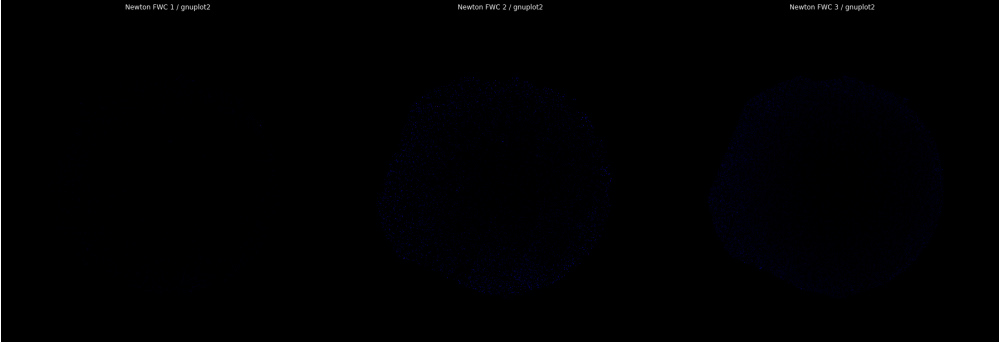
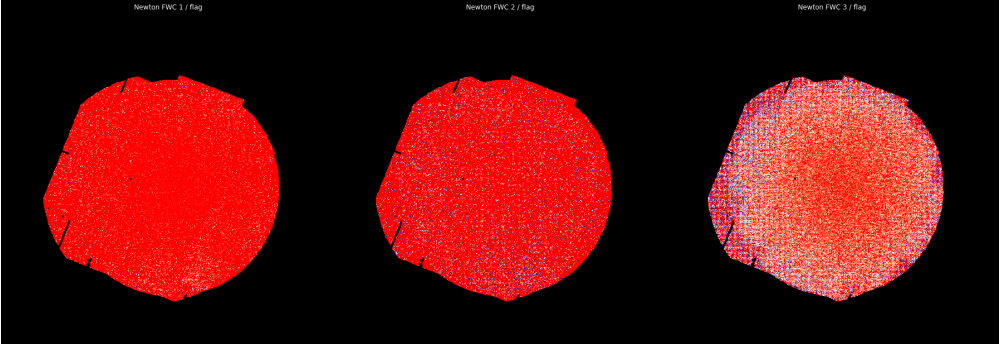
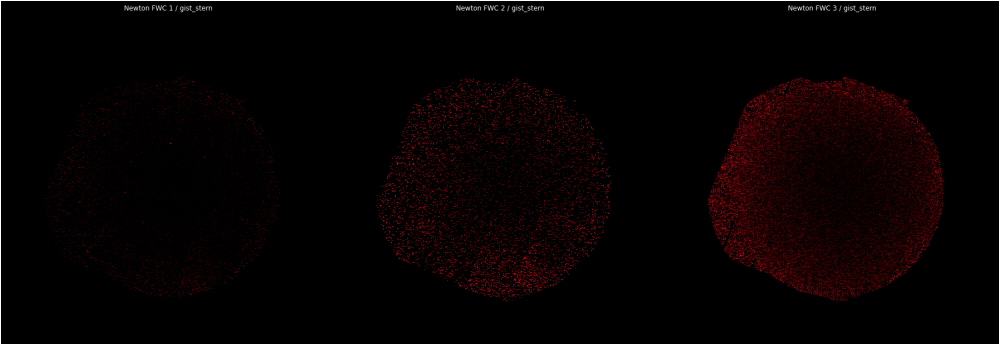
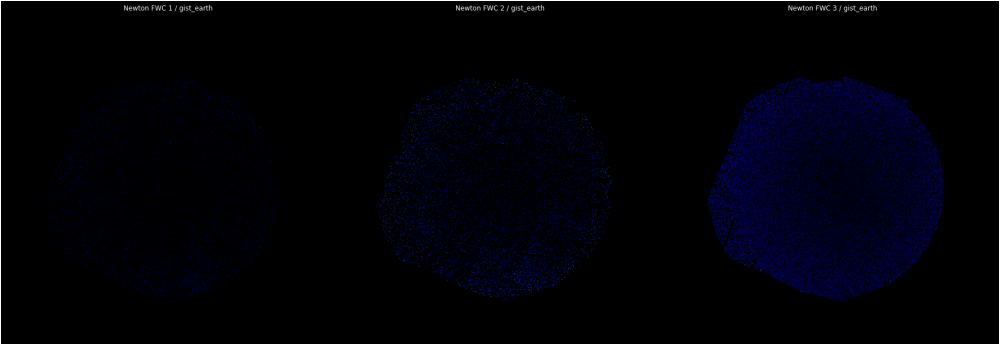
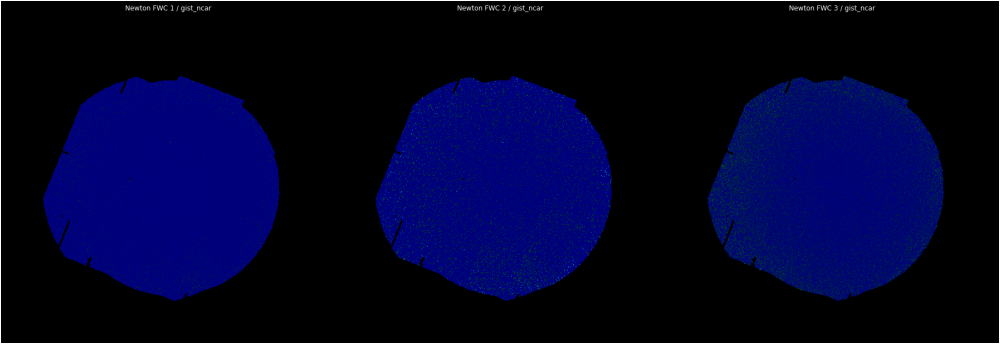
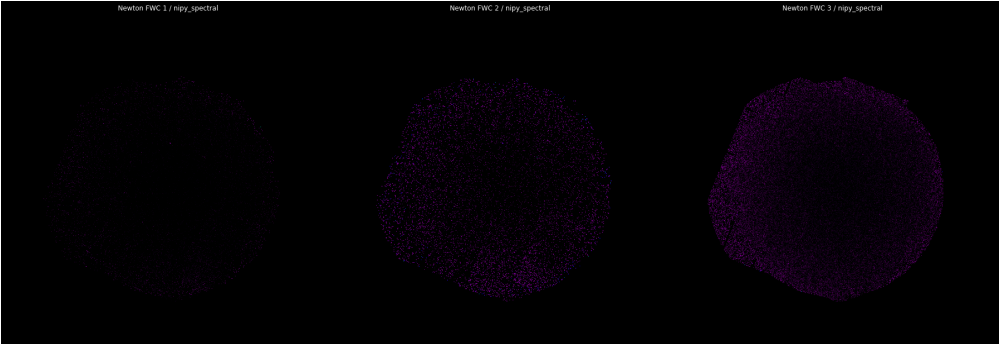
    axis[0].imshow(NEWTON_FWC[0] / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Newton FWC 1" + " / " + x_spec)
    axis[0].axis("off")

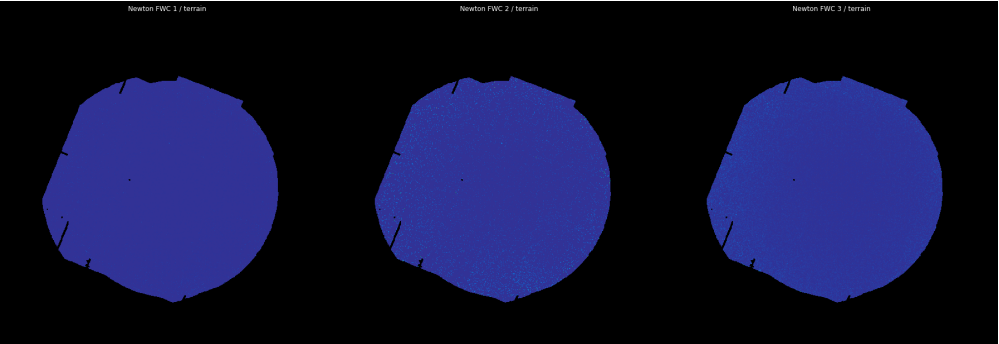
    axis[1].imshow(NEWTON_FWC[1] / 255.,cmap=x_spec)
    axis[1].set_title("Newton FWC 2" + " / " + x_spec)
    axis[1].axis("off")

    axis[2].imshow(NEWTON_FWC[2] / 255.,cmap=x_spec)
    axis[2].set_title("Newton FWC 3" + " / " + x_spec)
    axis[2].axis("off")

    plt.tight_layout()
    plt.show()
```







In []:

```
# XMM NEWTON OOT IMAGES
for x_spec in SPECTRAL_LIST:

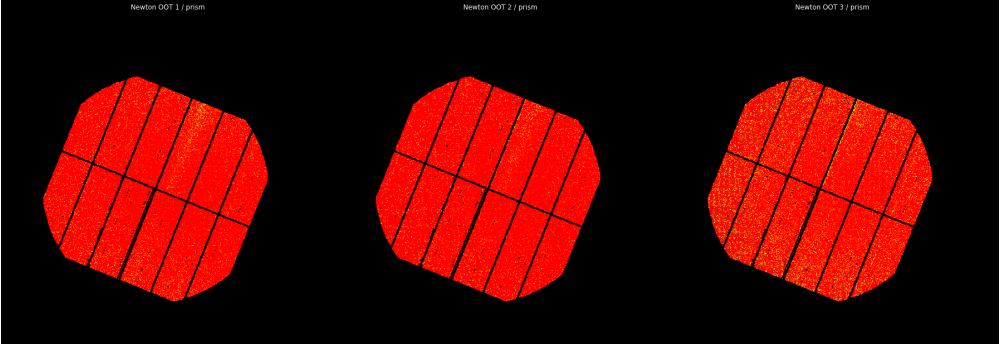
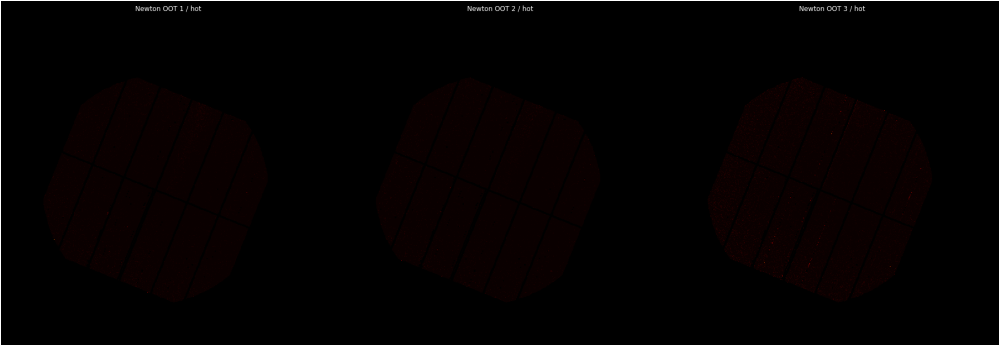
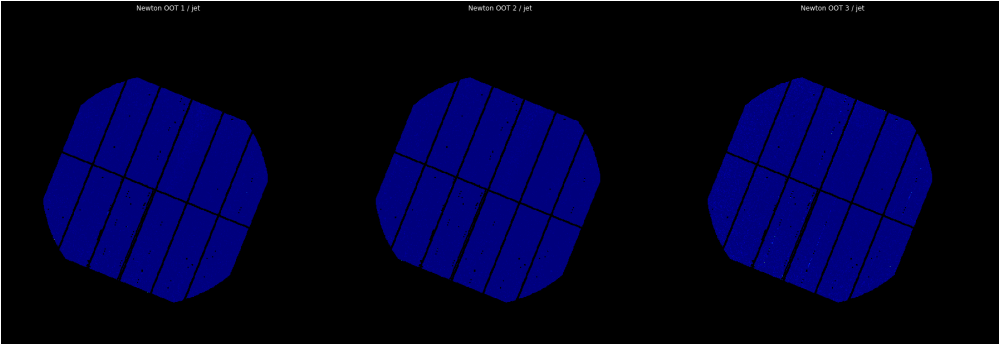
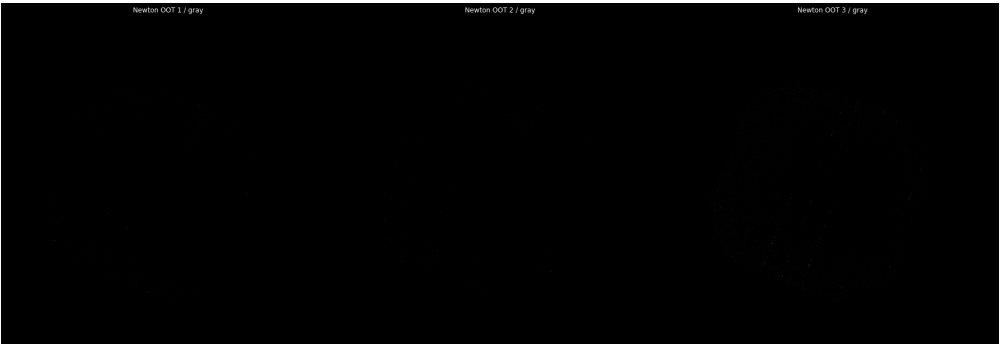
    figure,axis = plt.subplots(1,3,figsize=(25,25))

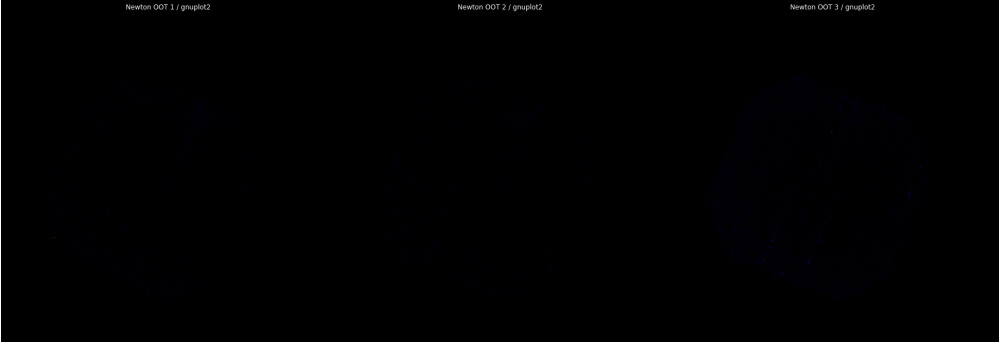
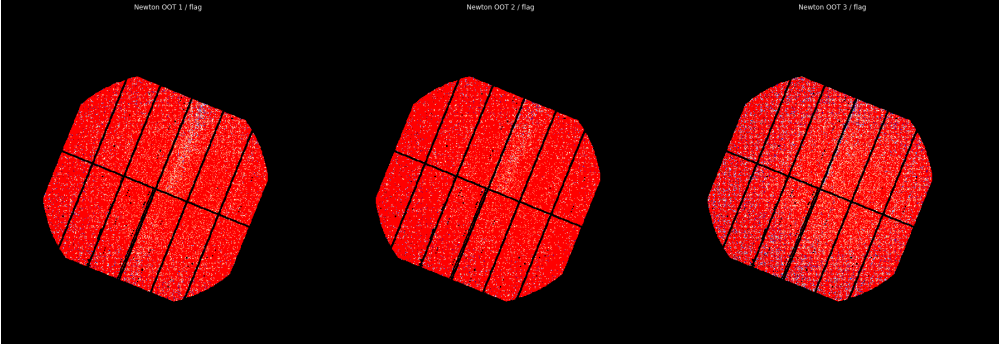
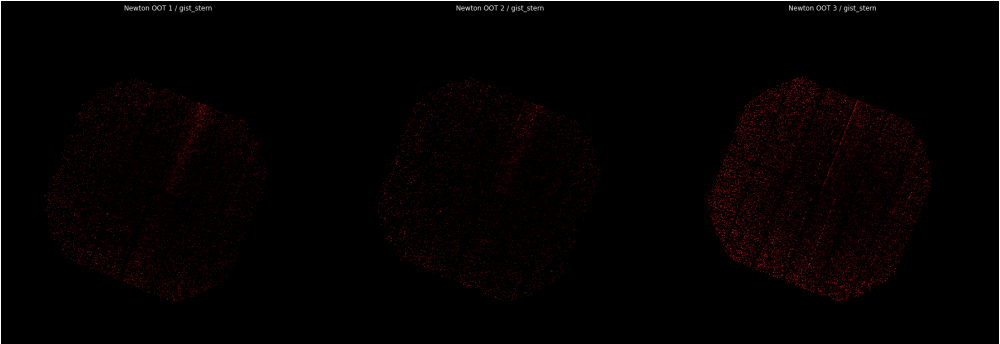
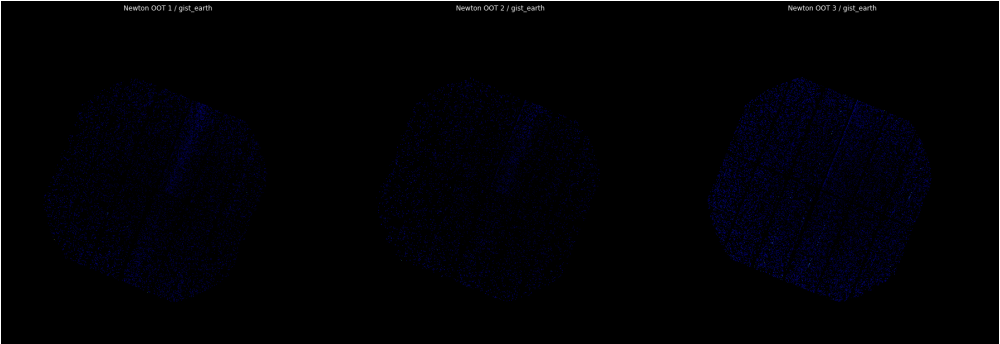
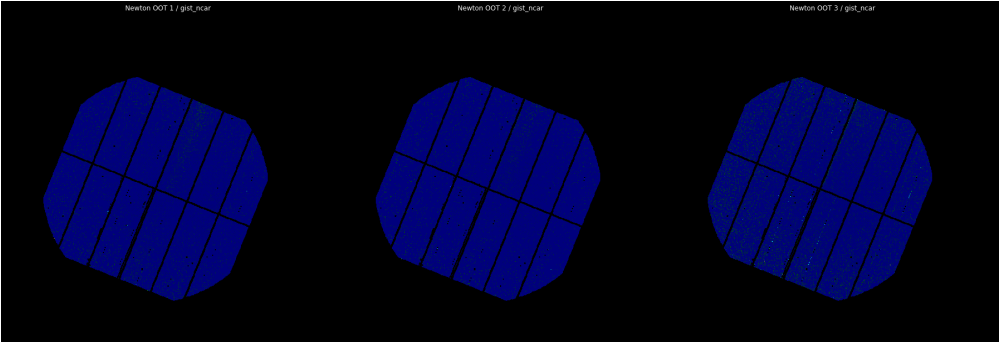
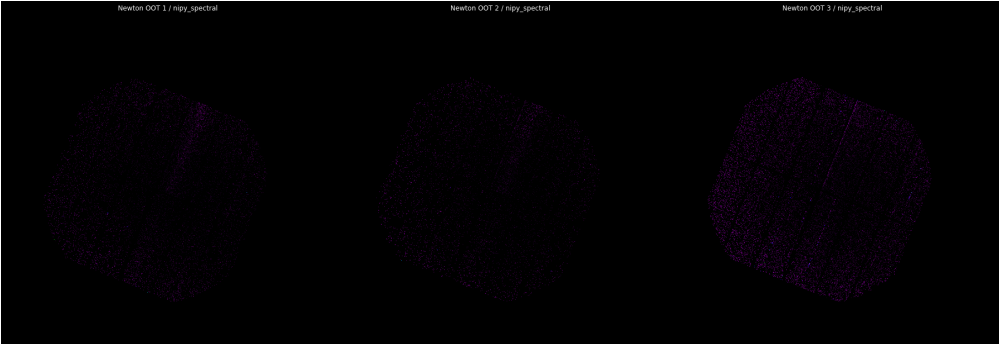
    axis[0].imshow(NEWTON_OOT[0] / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Newton OOT 1" + " / " + x_spec)
    axis[0].axis("off")

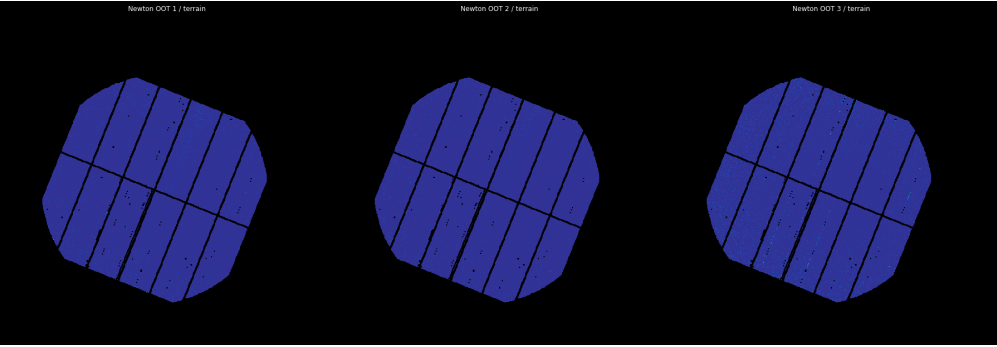
    axis[1].imshow(NEWTON_OOT[1] / 255.,cmap=x_spec)
    axis[1].set_title("Newton OOT 2" + " / " + x_spec)
    axis[1].axis("off")

    axis[2].imshow(NEWTON_OOT[2] / 255.,cmap=x_spec)
    axis[2].set_title("Newton OOT 3" + " / " + x_spec)
    axis[2].axis("off")

plt.tight_layout()
plt.show()
```







```
In [ ]: # Extraxting data that gives the most insight from the data above

WIDE_RANGE_XMM = NEWTON_RAW[1][200:480,200:490]
WIDE_RANGE_NI = NI_SCI[60:200,60:290]
WIDE_RANGE_HST_SCI = HST_SCI[380:550,320:490]
NUCLEUS_HST_SCI = HST_SCI[450:490,380:410]
```

```
In [ ]: # Applying filters on extracted data

# SOBEL Filter
SOBEL_WIDE_XMM = filters.sobel(WIDE_RANGE_XMM)
SOBEL_WIDE_NI = filters.sobel(WIDE_RANGE_NI)
SOBEL_WIDE_HST = filters.sobel(WIDE_RANGE_HST_SCI)
SOBEL_NUCLEUS_HST = filters.sobel(NUCLEUS_HST_SCI)
```

```
In [ ]: # Viewing filtered data

for x_spec in SPECTRAL_LIST:

    figure,axis = plt.subplots(1,4,figsize=(25,25))

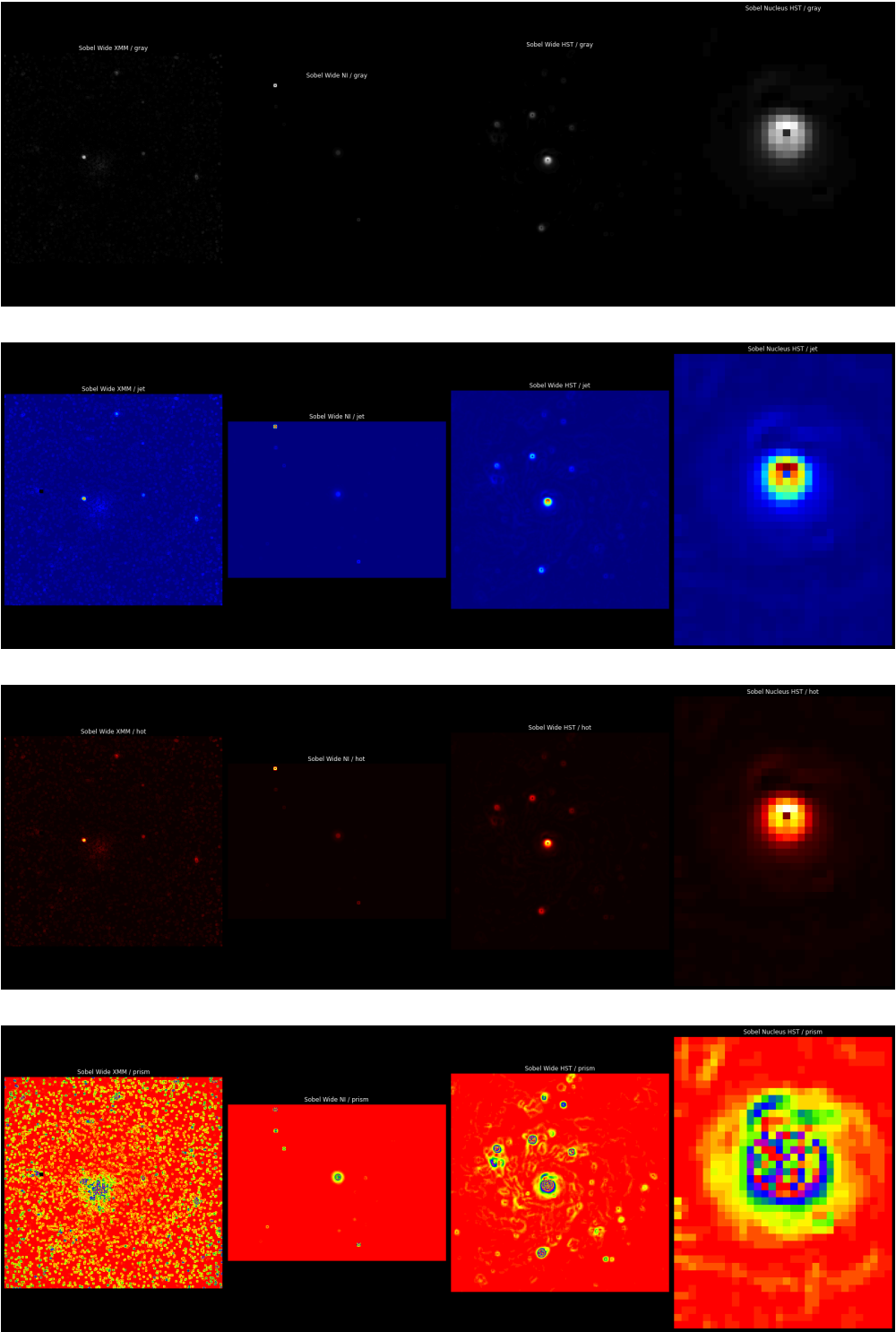
    axis[0].imshow(SOBEL_WIDE_XMM / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Sobel Wide XMM" + " / " + x_spec)
    axis[0].axis("off")

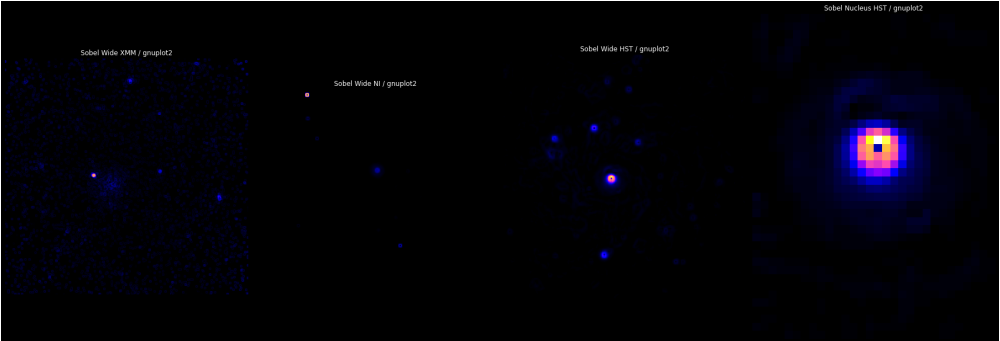
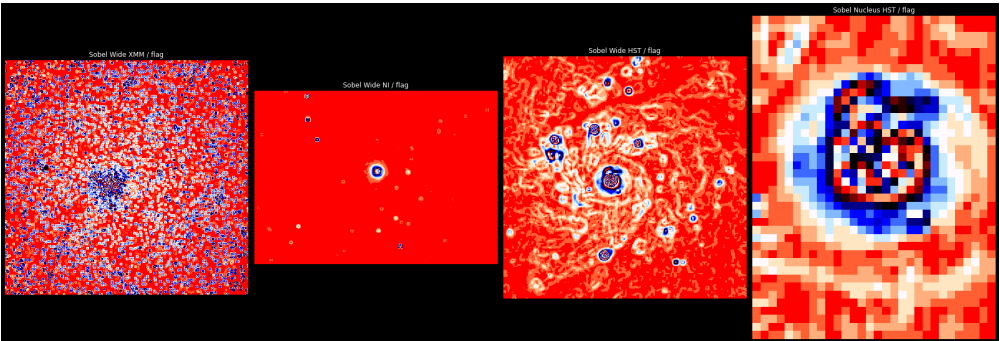
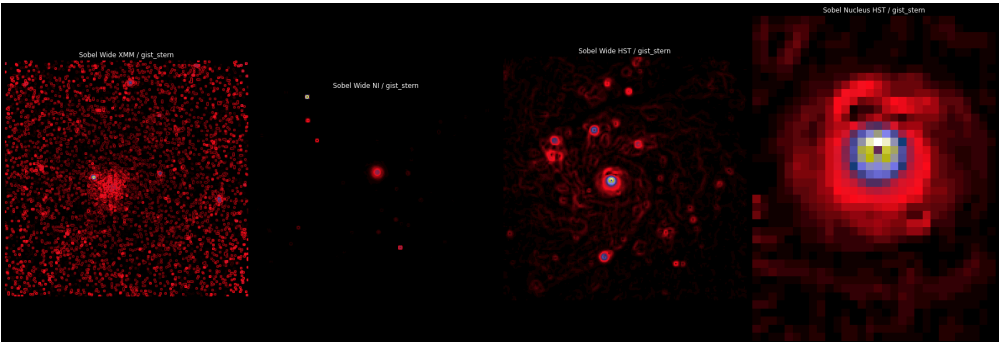
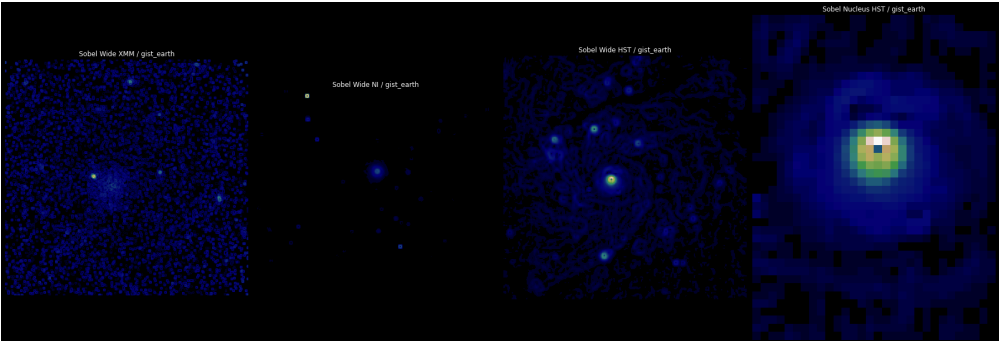
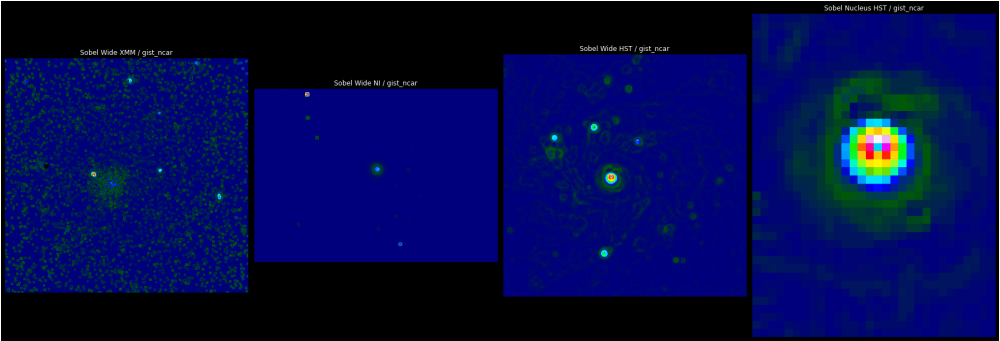
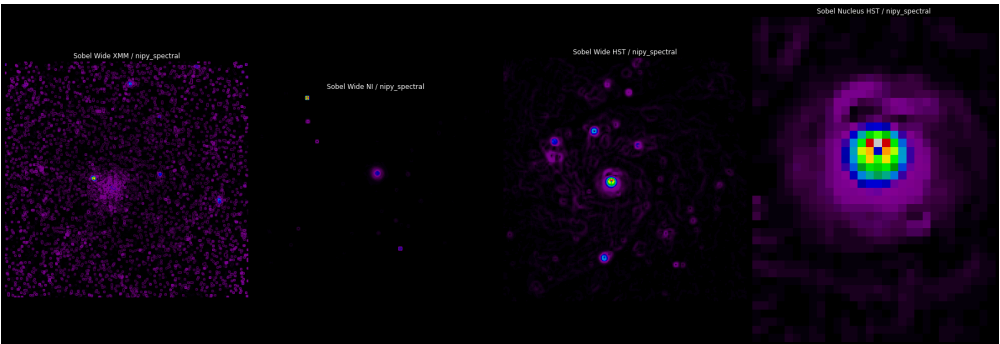
    axis[1].imshow(SOBEL_WIDE_NI / 255.,cmap=x_spec)
    axis[1].set_title("Sobel Wide NI" + " / " + x_spec)
    axis[1].axis("off")

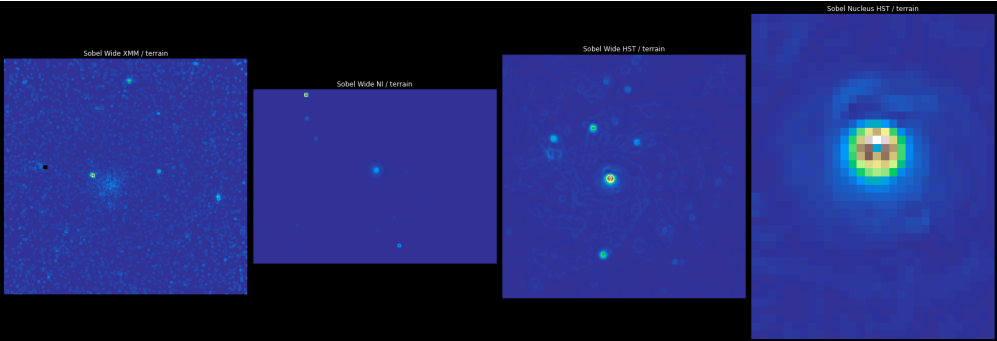
    axis[2].imshow(SOBEL_WIDE_HST / 255.,cmap=x_spec)
    axis[2].set_title("Sobel Wide HST" + " / " + x_spec)
    axis[2].axis("off")

    axis[3].imshow(SOBEL_NUCLEUS_HST / 255.,cmap=x_spec)
    axis[3].set_title("Sobel Nucleus HST" + " / " + x_spec)
    axis[3].axis("off")

    plt.tight_layout()
    plt.show()
```







```
In [ ]: # MEDIAN FILTER

MED_WIDE_XMM = signal.medfilt2d(WIDE_RANGE_XMM, kernel_size=(7,7))
MED_WIDE_NI = signal.medfilt2d(WIDE_RANGE_NI, kernel_size=(7,7))
MED_WIDE_HST = signal.medfilt2d(WIDE_RANGE_HST_SCI, kernel_size=(7,7))
MED_NUCLEUS_HST = signal.medfilt2d(NUCLEUS_HST_SCI, kernel_size=(7,7))
```

```
In [ ]: # Viewing filtered data

for x_spec in SPECTRAL_LIST:

    figure,axis = plt.subplots(1,4,figsize=(25,25))

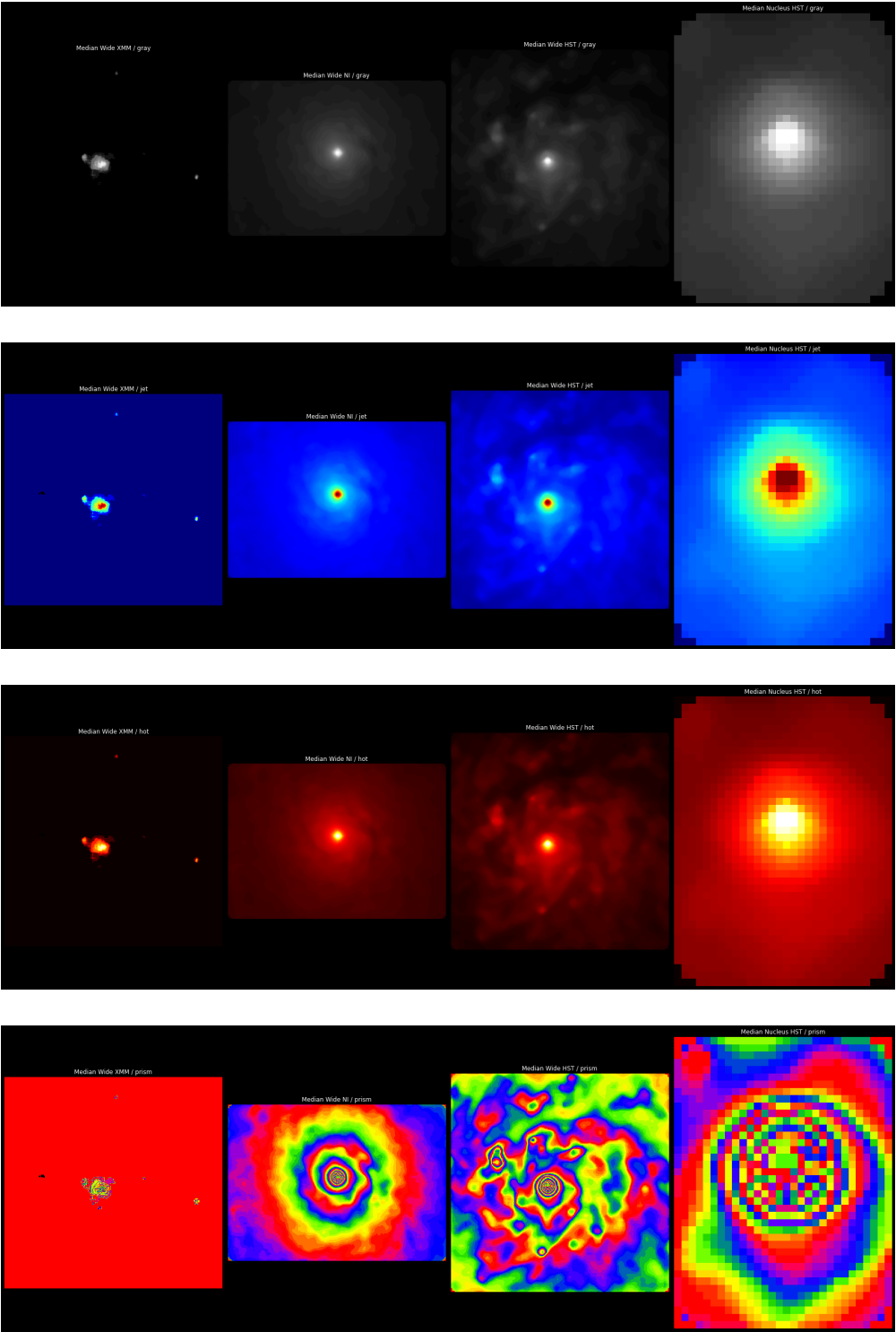
    axis[0].imshow(MED_WIDE_XMM / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("Median Wide XMM" + " / "+ x_spec)
    axis[0].axis("off")

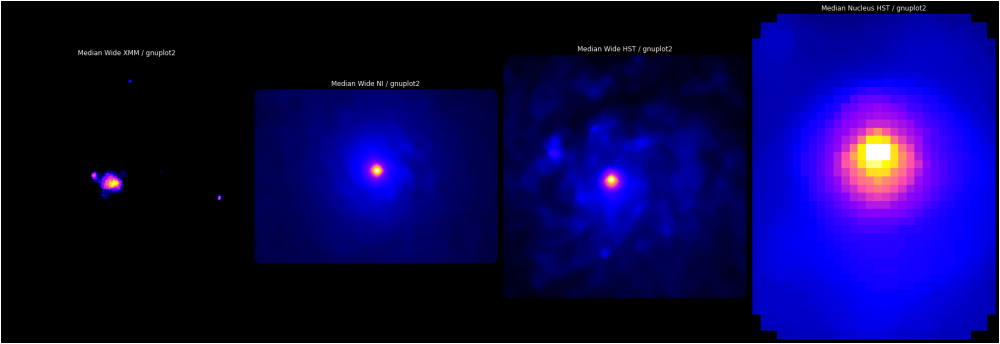
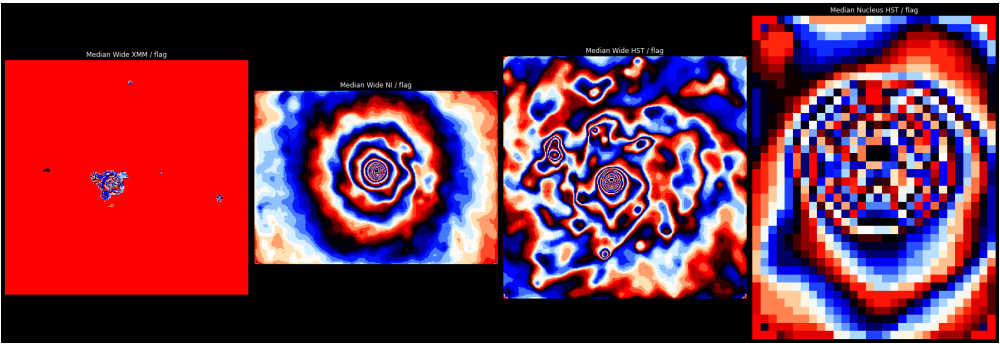
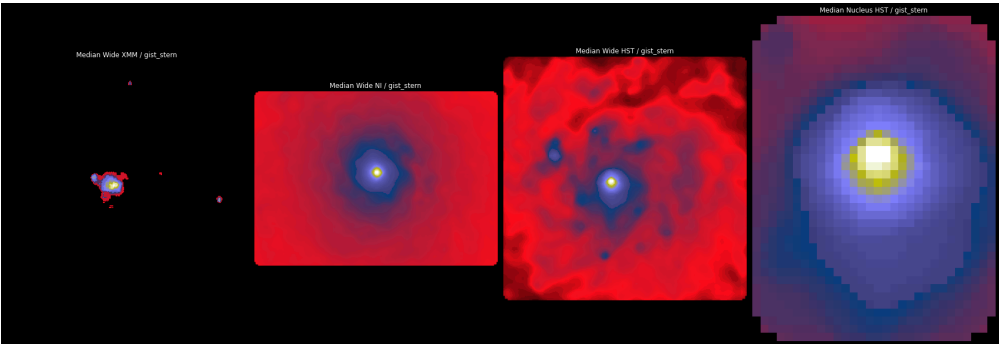
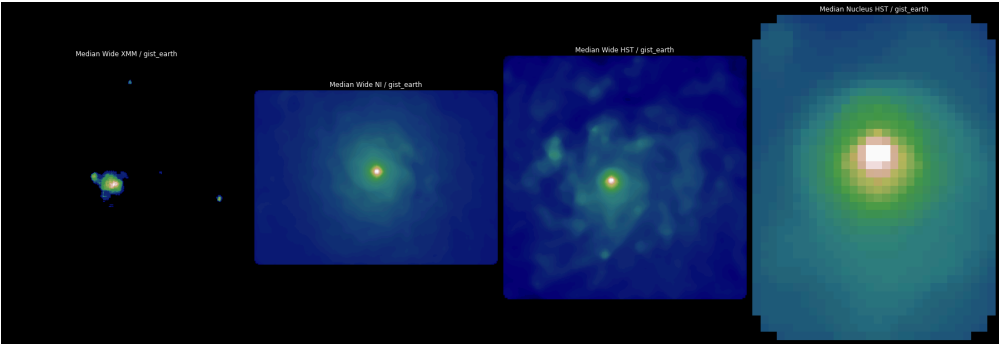
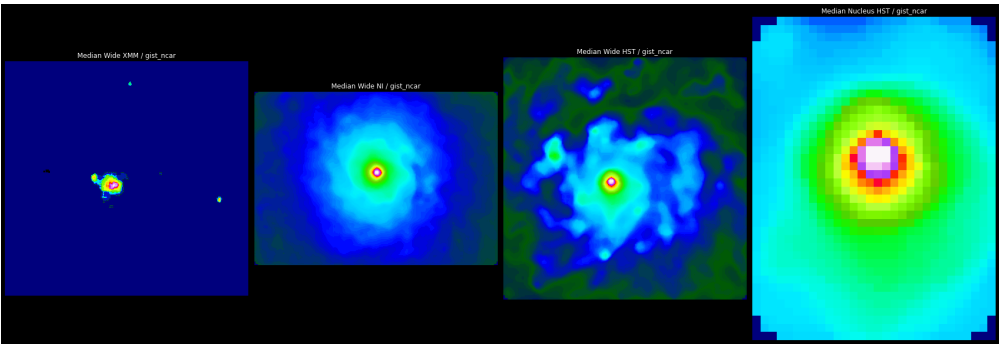
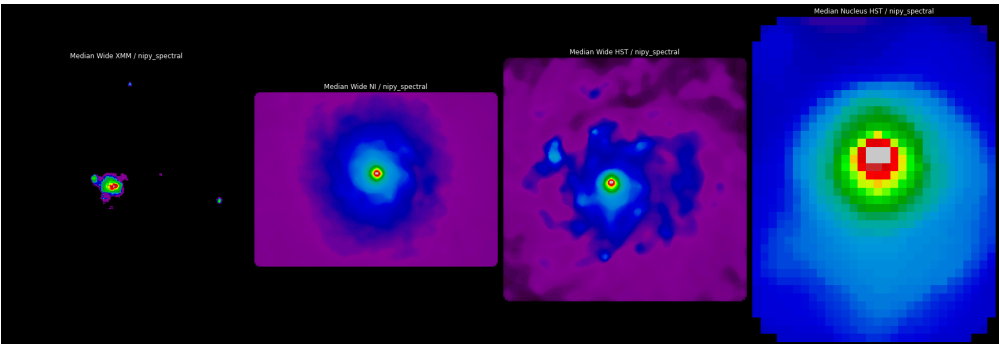
    axis[1].imshow(MED_WIDE_NI / 255.,cmap=x_spec)
    axis[1].set_title("Median Wide NI" + " / "+ x_spec)
    axis[1].axis("off")

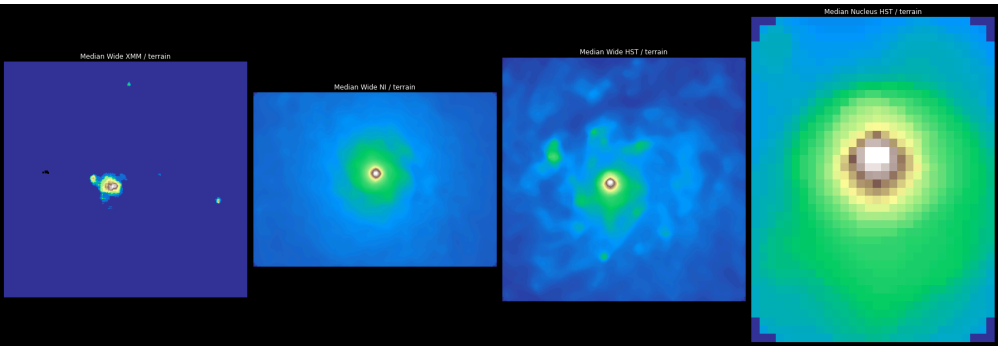
    axis[2].imshow(MED_WIDE_HST / 255.,cmap=x_spec)
    axis[2].set_title("Median Wide HST" + " / "+ x_spec)
    axis[2].axis("off")

    axis[3].imshow(MED_NUCLEUS_HST / 255.,cmap=x_spec)
    axis[3].set_title("Median Nucleus HST" + " / "+ x_spec)
    axis[3].axis("off")

    plt.tight_layout()
    plt.show()
```







INFERENCE

- * By reading the FITS data and viewing the images in 11 different color maps we were able to obtain 3 images that gave the most insight. They are as follows :-
RAW image from XMM sensor
SCI image from Near Infrared sensor
SCI image from HST sensor
- * We then widened these images and also got a nucleus image of HST thus we got total of 4 final images to perform filtering on.
- * Furthermore we applied SOBEL filter & MEDIAN filter on these chosen images which helps us detect edges in the images so that we can find a figure having close resemblance to a BLACK HOLE.
- * Finally we view these filtered images in 11 different color maps to get a better and more precise judgment of the galaxy center. The color maps we used are :-
["gray", "jet", "hot", "prism", "nipy_spectral", "gist_ncar",
"gist_earth", "gist_stern", "flag", "gnuplot2", "terrain"]

CONCLUSION

- After processing the images with 2 filters and then viewing them in 11 different color maps we were able to find images that can give us the most insight; these images are :-
- * Nucleus HST image with "sobel" filter in "hot" color map
 - * Nucleus HST image with "sobel" filter in "gnuplot2" color map
 - * Nucleus HST image with "Median" filter in "gnuplot2" color map
- To obtain better results we will perform feature detection on these images.

Step 4

To prove your hypothesis and have better insights on data, apply relevant feature detection algorithm, atleast 2 on the filtered data.

Write 3 points of inference, one-one on each feature. Last point as a conclusion, what you understood from it.

```
In [ ]: # Performing Feature detection on the images

# SATO filter on nucleus image

SATO_SOBEL_NUCLEUS = filters.sato(SOBEL_NUCLEUS_HST)
SATO_MED_NUCLEUS = filters.sato(MED_NUCLEUS_HST)
```

```
In [ ]: # Viewing SATO filter images

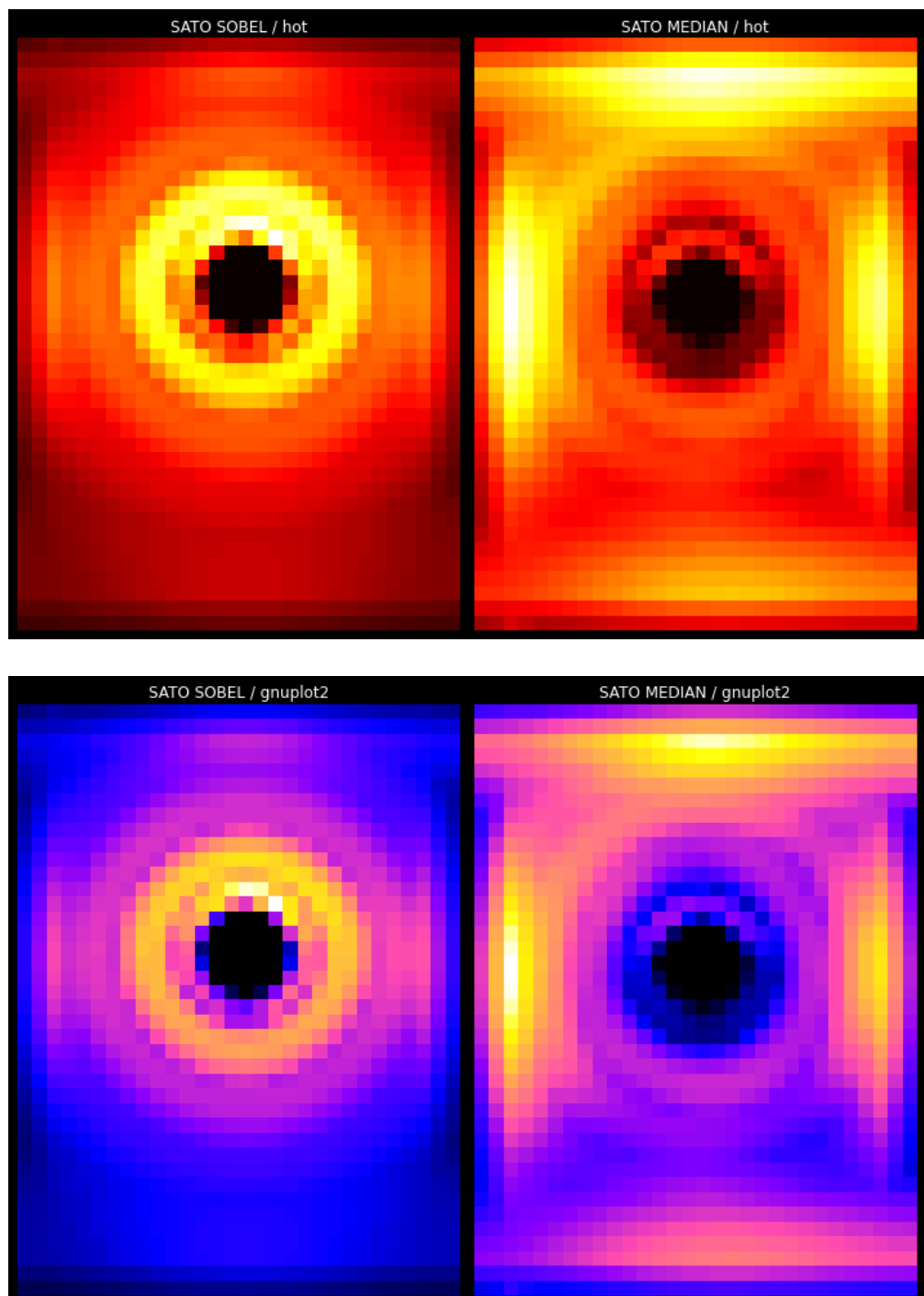
SPECTRAL_LIST_2 = ["hot","gnuplot2"]
for x_spec in SPECTRAL_LIST_2:

    figure,axis = plt.subplots(1,2,figsize=(10,10))

    axis[0].imshow(SATO_SOBEL_NUCLEUS / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("SATO SOBEL" + " / " + x_spec)
    axis[0].axis("off")

    axis[1].imshow(SATO_MED_NUCLEUS / 255.,cmap=x_spec)
    axis[1].set_title("SATO MEDIAN" + " / " + x_spec)
    axis[1].axis("off")

    plt.tight_layout()
    plt.show()
```



```
In [ ]: # MEIJERING filter on nucleus image

MEIJERING_SOBEL_NUCLEUS =
filters.meijering(SOBEL_NUCLEUS_HST,mode="nearest")
MEIJERING_MED_NUCLEUS = filters.meijering(MED_NUCLEUS_HST,mode="nearest")
```

```
In [ ]: # Viewing MEIJERING filter images

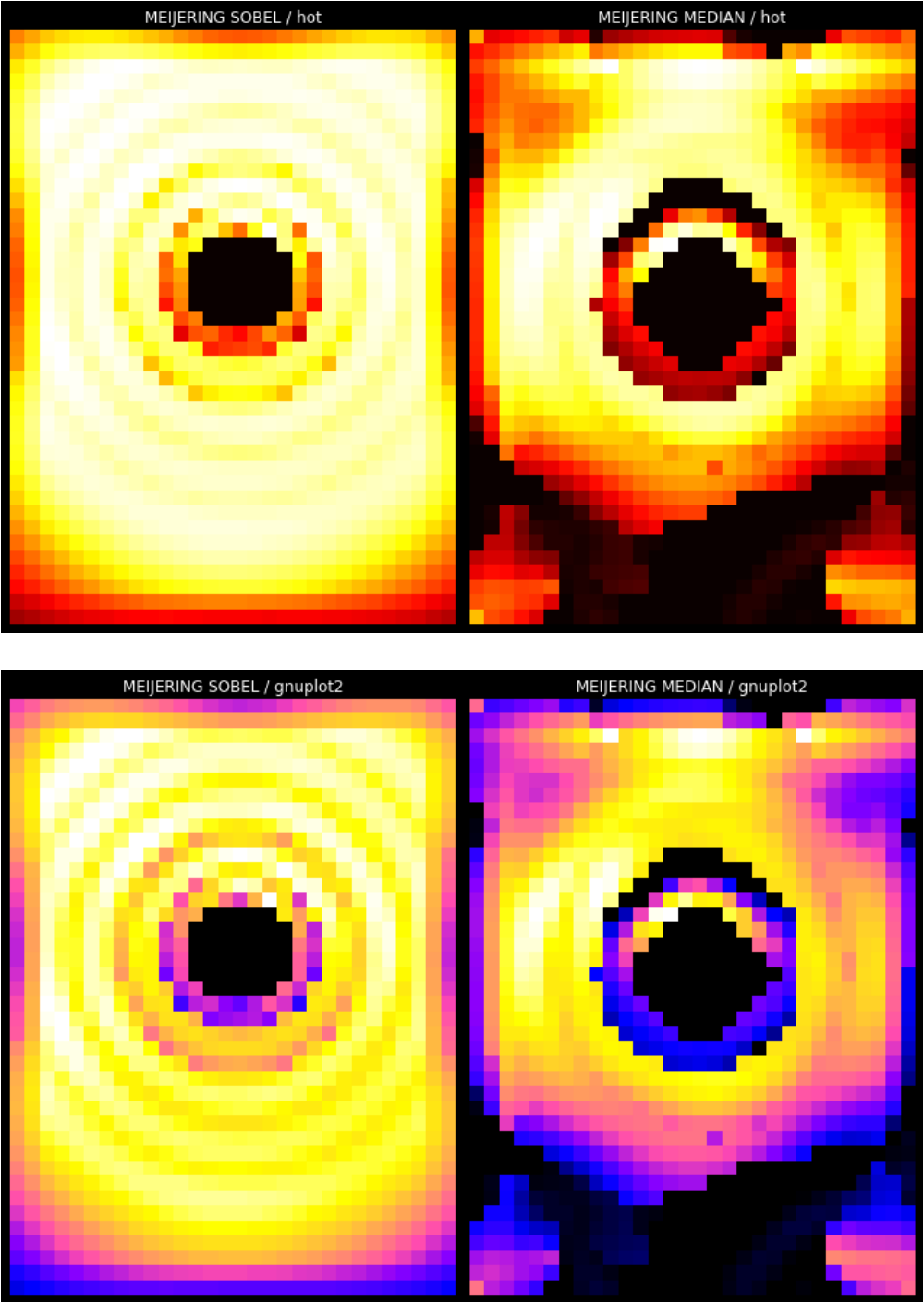
for x_spec in SPECTRAL_LIST_2:

    figure,axis = plt.subplots(1,2,figsize=(10,10))

    axis[0].imshow(MEIJERING_SOBEL_NUCLEUS / 255.,cmap=x_spec) #TARGET
    axis[0].set_title("MEIJERING SOBEL" + " / "+ x_spec)
    axis[0].axis("off")

    axis[1].imshow(MEIJERING_MED_NUCLEUS / 255.,cmap=x_spec)
    axis[1].set_title("MEIJERING MEDIAN" + " / "+ x_spec)
    axis[1].axis("off")

    plt.tight_layout()
    plt.show()
```



```
In [ ]: # Setting parameters for viewing WAVE PLOT

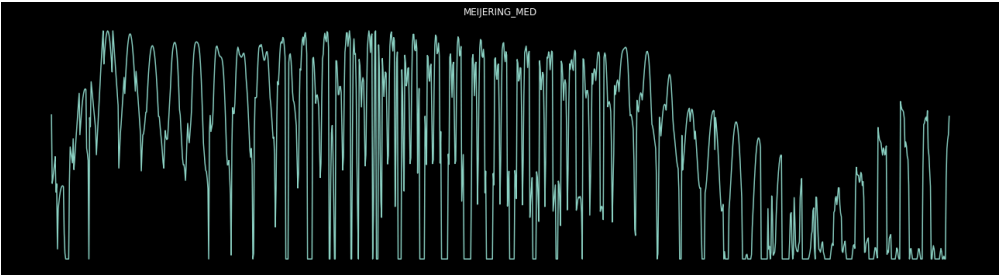
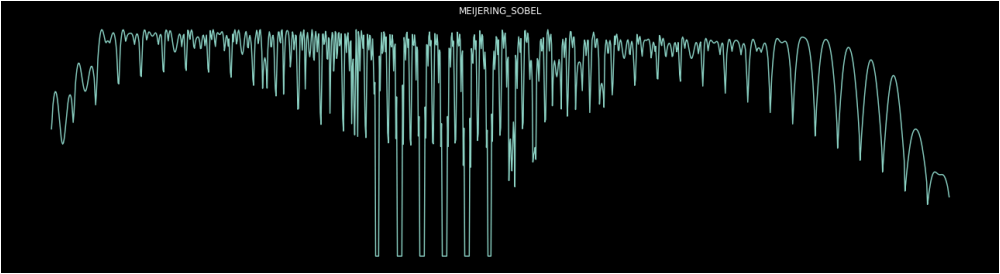
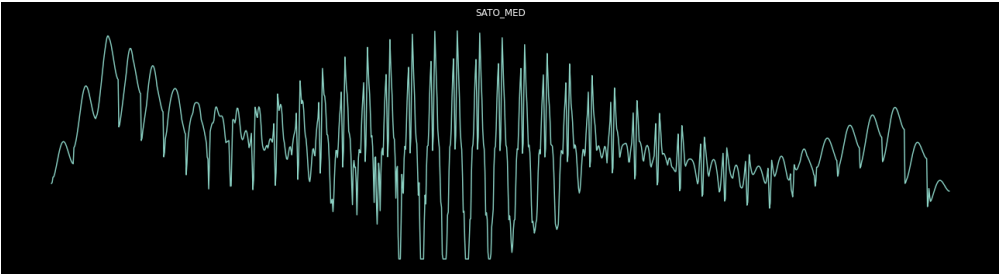
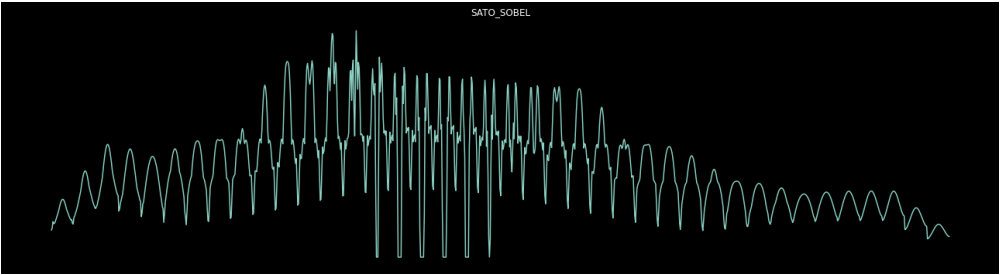
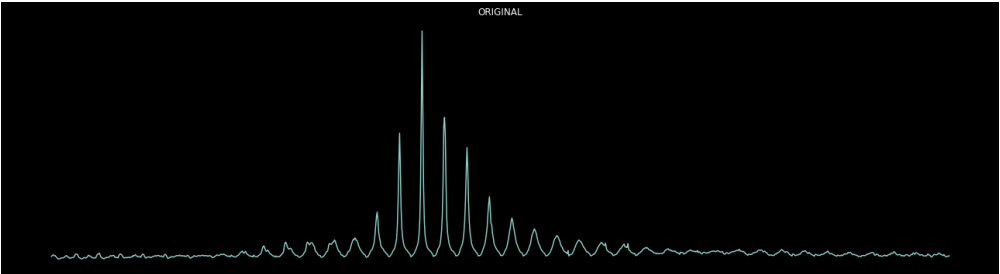
FORMATTING_LIST =
[NUCLEUS_HST_SCI,SATO_SOBEL_NUCLEUS,SATO_MED_NUCLEUS,MEIJERING_SOBEL_NUCLEU
S,MEIJERING_MED_NUCLEUS]
FORMATTING_LIST_FLATTEN =
[NUCLEUS_HST_SCI.flatten(),SATO_SOBEL_NUCLEUS.flatten(),SATO_MED_NUCLEUS.fl
atten(),MEIJERING_SOBEL_NUCLEUS.flatten(),MEIJERING_MED_NUCLEUS.flatten()]
FORMATTING_NAMES =
["ORIGINAL","SATO_SOBEL","SATO_MED","MEIJERING_SOBEL","MEIJERING_MED"]
```

```
In [ ]: plt.style.use("dark_background")
```

```
In [ ]: # Viewing WAVE PLOT
for x_formatting_type, x_formatting_name in
zip(FORMATting_LIST_FLATTEN,FORMATting_NAMES):

    plt.figure(figsize=(18,5))
    plt.axis("off")
    plt.plot(x_formatting_type)
    plt.title(x_formatting_name)

    plt.tight_layout()
    plt.show()
```



```
In [ ]: # Comparing WAVE PLOT with respective images

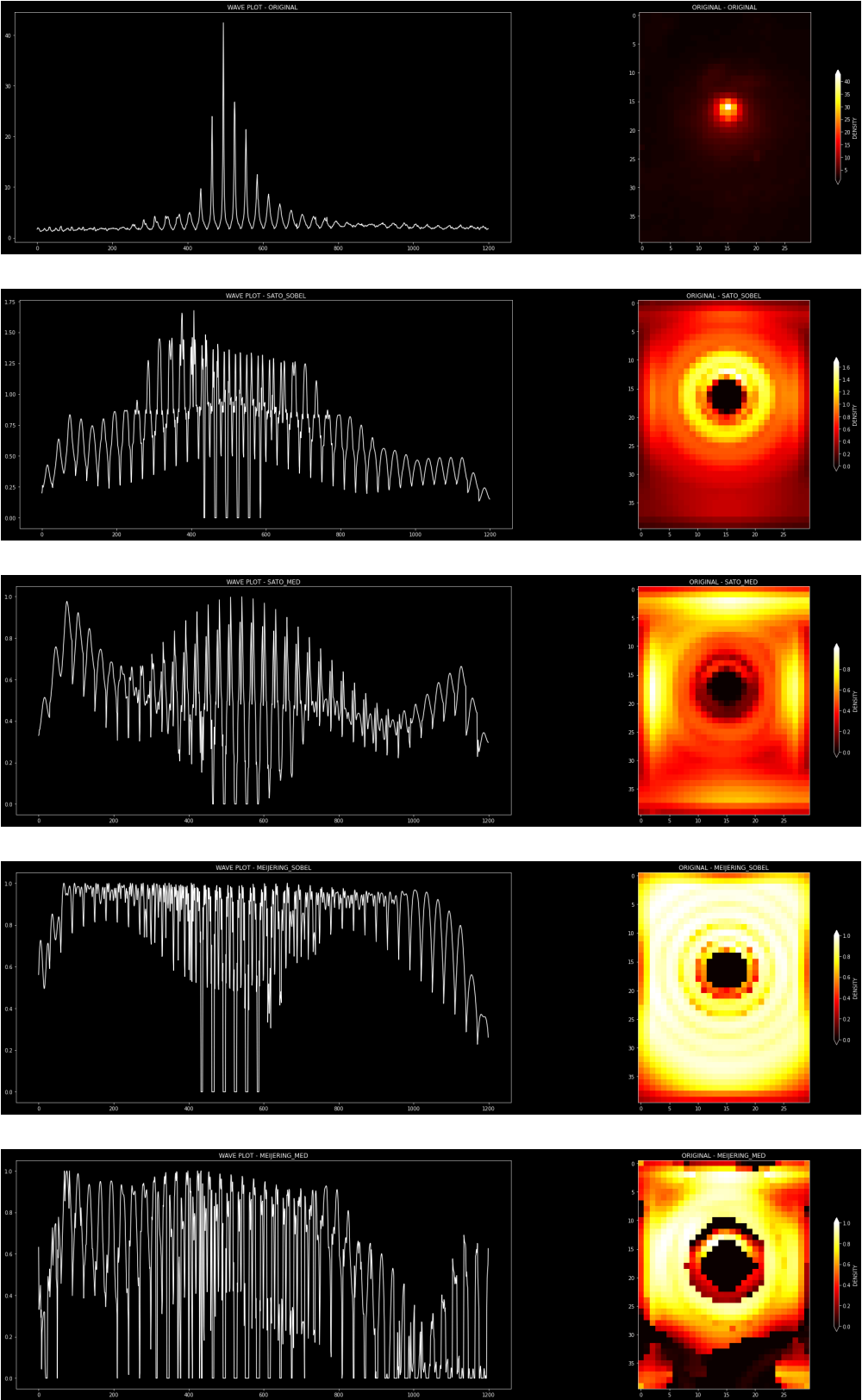
for x_formatting_type, x_formatting_name, x_formatting_original in
zip(FORMATting_LIST_FLATTEN,FORMATting_NAMES,FORMATting_LIST):

    figure,axis = plt.subplots(1,2,figsize=(25,7))

    axis[0].plot(x_formatting_type,color="white")
    axis[0].set_title(f"WAVE PLOT - {x_formatting_name}")

    DENSITY_FUNC = axis[1].imshow(x_formatting_original,cmap="hot")
    axis[1].set_title(f"ORIGINAL - {x_formatting_name}")

    figure.colorbar(DENSITY_FUNC,shrink=0.5,label="DENSITY",extend="both")
    plt.tight_layout()
    plt.show()
```



INFERENCE

* Here we applied SATO filter for feature detection on SOBEL & MEDIAN filtered images. SATO filter is used for corner detection in images. After applying SATO filter we can see a figure appearance similar to a context of a BLACK HOLE.

* Then we applied MEIJERING filter on the filtered images. MEIJERING filter is used to detect big gaps present in the image. After applying MEIJERING filter we were able to see pattern resemblance to BLACK HOLE.

* To give our research a bit more assurance we made a wave plot of all the filterd images and studied them next to the original images.

CONCLUSION

After extracting useful images from FITS data and filtering them with multiple filters & then after comparing those images with their flattened wave plot we were able to conclude that an figure resembling that to a black hole is seen in the nucleus photo of SCI image of HST sensor.
